
AxoSim: Learned Neuron Models for Fast Population Simulation and Inference-Time Adaptation

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Abstract

Detailed neuron simulators resolve morphology and ion-channel mechanisms, but repeated numerical integration is costly at population scale. Learned surrogates reduce this cost; in recurrent circuits, however, each prediction also affects the model’s future inputs. AxoSim combines morphology-conditioned GRU and Mamba neuron models with delayed synaptic routing, mutable efficacies, and differentiable population state. Under the standard AxoBench evaluation, AxoSim–GRU Medium reaches spike Mean F1 0.522, compared with up to 0.077 for released Branch-ELM models, and reduces Voltage and Dynamics SERA errors by 21.3% and 4.4%, while processing 16.8 million neuron-steps per second in the matched inference benchmark. On an RTX 5090, the runtime simulates 65,536 neurons with 4,000 retained contacts per neuron at $3.38\times$ real-time speed. On the same consumer-grade GPU, populations of 20,000 neurons are trained in real time with full backpropagation through time (BPTT) at 2,000 retained contacts per neuron. In a temporal-shift experiment on Spiking Heidelberg Digits (SHD), morphology and synaptic adaptation raise target accuracy by 6.46 ± 1.20 percentage points across six independently trained models. A separate corpus provides 50,000 validated population traces from 100 recurrent connectomes with connectome-disjoint splits. Together, these experiments separate teacher-forced fidelity from recurrent simulation and inference-time adaptation.

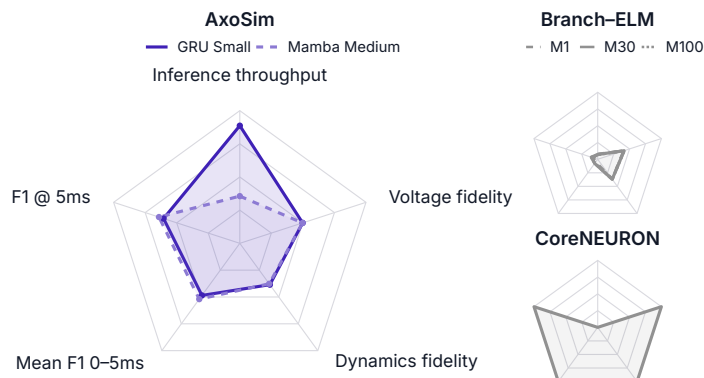


Figure 1: Five quantities available for all three systems. AxoSim shows Mamba Medium and the throughput-leading GRU Small alongside three released Branch-ELM sizes; CoreNEURON’s self-comparison defines perfect fidelity. Raw values, mappings, and runtime contracts appear in Appendix A.11.

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1 Overview

1.1 The simulation problem

Biophysically detailed neuron simulators integrate membrane mechanisms over a branched morphology, which preserves a direct relation between model parameters and cellular processes. That detail is indispensable when the question concerns a named conductance, compartment, or synaptic mechanism, but it also makes repeated simulation and gradient-based fitting expensive as the number of cells grows. Learned neuron models instead fit responses generated by a detailed teacher and approximate the map from synaptic input histories to somatic voltage and spikes [2, 19].

Most evaluations of learned neurons hold the input trace fixed while comparing the predicted and teacher responses. This teacher-forced evaluation measures the quality of a single response. A recurrent population introduces an additional difficulty because a predicted spike is delayed, delivered to downstream dendrites, and incorporated into the input for later predictions. An error can therefore change the current output and the distribution of future inputs. Population simulation must measure this feedback directly because single-neuron accuracy omits its recurrent consequences.

AxoSim is a learned-neuron simulation platform designed around that distinction. It complements the mechanistic teacher with compatible interfaces for fitting neuron responses, connecting the fitted models through explicit synaptic contacts, advancing persistent populations, and differentiating through a supplied trajectory. Its learned coordinates are predictive variables rather than identified ion-channel mechanisms. CoreNEURON simulations of Hay layer-5 pyramidal cells provide the principal detailed-neuron targets, while released Branch-ELM models provide the principal learned reference.

1.2 From single neurons to recurrent populations

At the single-neuron level, each AxoSim model receives events that preserve their sign and dendritic destination, combines them through a morphology-conditioned front end, and predicts a somatic-voltage coordinate and spike logit at every native millisecond. GRU and Mamba temporal cores share this input and output contract, so their fidelity and execution cost can be compared directly. The full models prioritize response fidelity. The compact AxoSim-Lite specialization preserves the same event semantics while reducing state and computation for persistent connected populations.

The population runtime maintains recurrent neuron state and a directed graph of synaptic contacts. Each contact defines its source, sign, branch destination, delay, and mutable efficacy. During each simulation step, thresholded model outputs define spikes. The runtime routes those events after their delays, accumulates their efficacies at the target branches, and records requested telemetry. A separate supplied-history computation differentiates through efficacy-weighted input aggregation and the complete temporal response. For large-population inference, fused GPU kernels organize connected execution in a phase- and tile-major layout.

Parameter ownership determines the scale of adaptation. Shared model weights define the learned response function; behavior parameters belong to individual neurons, morphology parameters are shared by a structural class, and synaptic parameters belong to individual contacts. Any selected group can receive gradients through the complete supplied sequence. This separation allows the same interface to express morphology transfer, changes in synaptic efficacy, and task adaptation while preserving their interpretation as predictive coordinates.

1.3 Evidence across fidelity, execution, and adaptation

AxoBench introduced spike Mean F1 across 0–5 ms matching tolerances, Voltage SERA, and Dynamics SERA as the core metric set for single-neuron evaluation [25]. Section 6 relates these three quantities to parameter count, forward-pass computation, and inference throughput. The largest separation from Branch-ELM occurs in spike timing. Voltage SERA also decreases substantially, whereas the smaller Dynamics-SERA change identifies native-time voltage increments as the harder quantity.

Paired synaptic interventions extend this comparison to changes in the teacher’s response, with both voltage-change error and condition-wise firing measured directly. Population timing then covers a complete recurrent step at explicit neuron and contact counts, while connected diagnostics measure activity after predictions enter the feedback loop. Adaptation analyses test transfer to excluded morphologies, a controlled conductance shift and a temporal distribution shift on SHD. Table 1 associates each result with its measurement condition.

Table 1: Principal AxoSim results and the conditions attached to each measurement.

Question	Result	Measured condition
Neuron fidelity	Mean F1 0.522; the lowest AxoSim Voltage SERA is 21.3% below its Branch-ELM reference, and the lowest Dynamics SERA is 4.4% below its reference.	Ordinary AxoBench, held-out morphology-balanced traces.
Intervention response	Response SERA is 9.57–10.70% below the strongest Branch-ELM response reference, with lower error in all twenty paired model comparisons.	350 scoring pairs across five input interventions and five morphologies; 50 separate baseline calibration pairs.
Population execution	1,048,576 neurons advance at 1.177 billion neuron-steps/s with 362 incoming synaptic contacts per neuron; 65,536 neurons with 4,000 contacts advance at 3.38 times real time.	Complete RTX 5090 step with delayed routing, mutable efficacies, state, and 1% telemetry.
Population data	50,000 detailed-teacher traces from 100 recurrent connectomes assigned to disjoint context splits.	Five excitatory morphologies and one inhibitory morphology; complete connectomes define the split.
Adaptation	SHD target accuracy improves by 6.46 ± 1.20 percentage points; 20,000 neurons with 2,000 retained contacts complete a 256-step full-BPTT update in real time.	Six independently trained SHD source models; complete full-BPTT update on one RTX 5090.
Biological transfer	A 58-value adapter improves spike and dynamics fidelity on five excluded morphologies; synaptic parameters recover one controlled conductance shift.	Predictive transfer is established; cellular identification requires further controlled interventions.

2 Morphology-conditioned neuron models

2.1 Dendritic integration and morphology

A learned neuron maps a history of synaptic events to somatic voltage and spikes. At native millisecond t , contact k contributes an event $x_{t,k}$ with a sign, dendritic destination, and efficacy. For morphology descriptor m , the model predicts a spike logit $\hat{\ell}_t$ and voltage coordinate $\hat{v}_t$,

$$(\hat{\ell}_t, \hat{v}_t) = f_\theta(x_{1:t}; m).$$

Voltage is converted to millivolts for evaluation, and a calibrated logit threshold determines spike events. Thus both outputs describe the response at the native time resolution.

The dendritic front end builds on Branch-ELM’s assignment of synapses to branches and integration of branch contributions before temporal memory [19]. AxoSim conditions synaptic gains on morphology, applies learned saturating transforms to branch currents, and projects the branch responses into a temporal representation. A morphology embedding and route-occupancy features additionally encode the cell identity and the usage and multiplicity of its synaptic routes. Consequently, the temporal core receives both the integrated event values and information about the structure through which those values arrived.

2.2 Temporal memory and response readouts

AxoSim-GRU integrates the branch representation with gated recurrent updates [3], while AxoSim-Mamba uses selective state-space blocks [7]. Both retain a persistent state during streaming inference. For complete input histories, the Mamba recurrence also permits a parallel scan with linear work and logarithmic parallel depth. Across model layers, the temporal transformation follows a pre-normalized residual path,

$$z_{\ell+1} = z_\ell + \gamma_\ell \mathcal{M}_\ell(\text{LN}(z_\ell)),$$

where z_ℓ is the layer input, $\mathcal{M}_\ell$ is the temporal transformation, and the learned channel-wise multiplier γ_ℓ scales its contribution. The identity path preserves a direct route across depth, following the residual-learning principle [9]; temporal optimization still depends on recurrent gradient propagation [14].

The Mamba family further refines the state-space response with causal convolutions at dilations 1, 4, and 16, so recent local events remain available alongside the recurrent memory. Separate linear heads then predict the voltage coordinate and spike logit. Their coupling is explicit, $\hat{\ell}_t = \ell_t^{\text{head}} + c_v \hat{v}_t$, with $c_v = 4$ in the reported full models. This joint readout links the spike prediction to the voltage trajectory while retaining a separately learned spike head.

Matched retraining of a two-layer Mamba configuration quantifies the value of these front-end choices. Removing route-occupancy conditioning increased Voltage Root-SERA by 0.423 mV and reduced the trapezoidal average of spike F1 over 0–5 ms by 0.087, with paired 95% intervals of [0.370, 0.477] and [0.072, 0.102], respectively. Removing the morphology embedding, branch saturation, local temporal refinement, or spike-voltage coupling also increased Voltage Root-SERA in that matched panel. These measurements establish the value of the components in that architecture; Appendix A.4 gives the training recipe and full results, distinguishing its Root-SERA and trapezoidal F1 from the current SERA and arithmetic Mean F1.

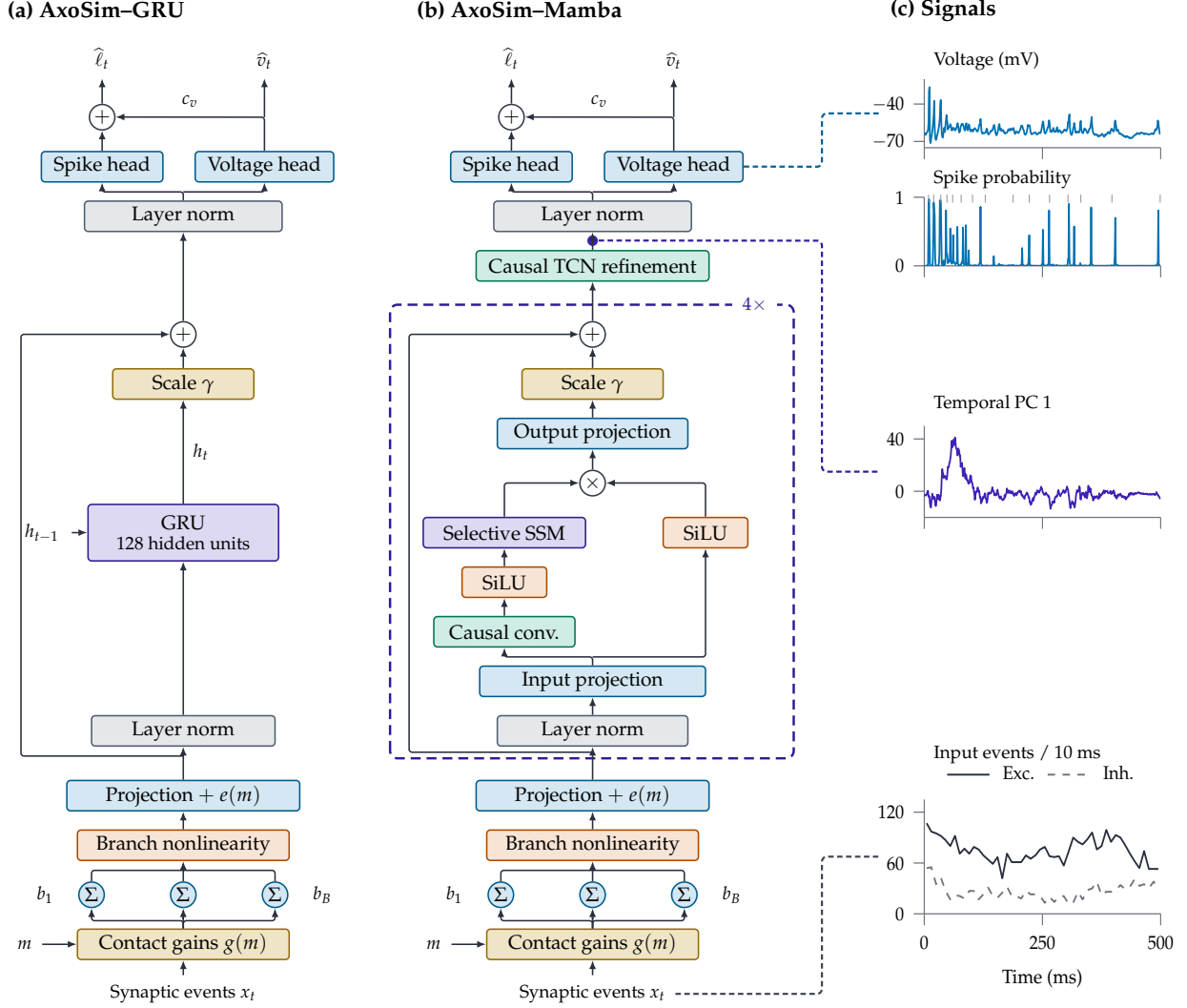


Figure 2: Neuron model architectures and a measured response. (a,b) Computation proceeds upward from morphology-conditioned branch aggregation to residual temporal updates and separate output heads. Branch routing and aggregation follow Branch-ELM. Mamba Medium repeats the illustrated block four times before causal temporal refinement; γ scales residual updates, and c_v couples voltage into the spike logit. (c) Dashed lines associate the Mamba stages with one held-out 500-ms trace. Outputs and temporal PC 1 are shown at 1-ms resolution; input events use 10-ms counts. Gray ticks mark detailed-simulator spikes.

3 Causal population execution

3.1 Four-step forecasts and delayed feedback

AxoSim-Lite uses a compact recurrence for persistent populations. One update integrates a completed four-millisecond block of branch inputs and predicts the spike logit and voltage at each millisecond of the next block. If τ is the block boundary and h_b is the persistent neuron state, the four forecasts satisfy

$$(\hat{y}_\tau, \hat{y}_{\tau+1}, \hat{y}_{\tau+2}, \hat{y}_{\tau+3}, h_{b+1}) = F_\theta(x_{\tau-4:\tau-1}, h_b).$$

Each $\hat{y}_t$ contains the two native-time outputs. The runtime uses these four predictions directly, retaining their individual event times.

The forecast horizon determines the minimum recurrent delay. For a horizon of P native steps, a newly predicted spike at $\tau + r$, with $0 \leq r < P$, reaches a target no earlier than $\tau + r + d_{\min}$. Requiring $d_{\min} \geq P$ therefore places that arrival at or beyond $\tau + P$, after the current forecast block (Figure 3). The target’s current prediction is consequently independent of newly predicted events inside that block. Delay-based lookahead is an established principle of parallel neural simulation [13]; here it schedules a learned four-step forecast with $P = 4$ and a minimum contact delay of 4 ms.

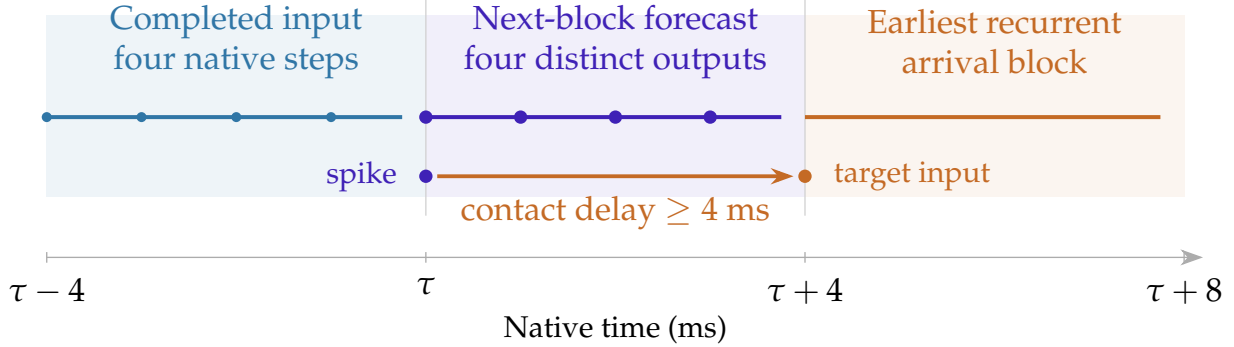


Figure 3: Causal scheduling of a four-step neuron forecast. At boundary τ , the completed input block supplies four distinct predictions. Even the earliest predicted spike, at τ , cannot reach another neuron before $\tau + 4$ under the minimum 4-ms contact delay. Later predicted spikes arrive correspondingly later.

Connected execution combines these forecasts with persistent neuron state and a delayed event ring. Thresholded spike logits define events; the router then accumulates their signed, efficacy-weighted contributions at destination branches and scheduled native times. Neuron phases are staggered across tiles to spread recurrent work, while the fused kernels operate on a phase- and tile-major layout. Requested trajectory samples are transferred asynchronously to the host. Thus the connected measurements include state updates, event formation, contact routing and telemetry.

3.2 Procedural contacts with mutable efficacies

In the structured population graphs, contact identity is reconstructed from neuron indices, source roles and channel offsets. The forward mapping determines the source, sign, destination branch and delay of each contact; its inverse enumerates the targets reached by an emitted event. This procedural representation replaces explicit source and destination index arrays while retaining a separate efficacy for every contact. Native delay-ring accumulation uses the reconstructed identity and the corresponding mutable efficacy, so updates to one contact alter that contact’s contribution without changing the topology. The representation applies to the declared typed, structured graphs; explicit contact indices remain available in the general tensor interface.

Storage then separates connectivity from neuron memory. With K incoming contacts, the deployed 8-bit efficacy bank requires K bytes per neuron, apart from a shared scale, whereas recurrent state and the delayed branch ring depend on state width, branch count and delay horizon. For example, 4,000 contact efficacies occupy 4,000 bytes per neuron; the eight-component BF16 recurrent state occupies 16 bytes. The remaining coefficient and ring storage, and the additional gradient and optimizer storage during adaptation, are specified in Appendix A.6. This separation makes contact density a measurable memory and routing tradeoff rather than a change in the number of shared neuron-model weights.

4 Compact parameters and temporal differentiation

4.1 Neuron, structural-class and contact parameters

Compact adaptation modifies input sensitivity, memory decay, the recurrent proposal and the output readout. Let u_b be the projected input block and h_b the persistent state. Token offsets and gains produce $\tilde{u}_b$; a decay coordinate d , proposal gain g_h and proposal offset o_h then modify the Lite transition,

$$h_{b+1} = d \odot h_b + (1 - d) \odot [(1 + g_h) \odot \tanh(W_u \tilde{u}_b + W_h h_b + b_h + o_h)].$$

Here W_u, W_h, b_h are shared projections and bias, and d is sigmoid-transformed. Output gains and offsets complete seven coefficient groups. For token width $D = 24$, state width $H = 8$, forecast horizon $P = 4$ and two outputs per native step, these groups contain $2D + 3H + 2P \cdot 2 = 88$ deployed coefficients.

The population interface assigns these coefficients at two scales (Table 2). A neuron-specific behavior bank permits individual specialization, while a morphology bank shares changes across one structural class. Their coefficients enter additively. A third bank assigns one log-efficacy $s_{i,k}$ to contact k of neuron i , whose positive efficacy $\exp(s_{i,k})$ multiplies the incoming event before branch integration. Selecting the trainable banks therefore determines whether adaptation changes individual responses, class-shared responses or synaptic input weights. Shared neuron-model weights remain separate from all three.

Table 2: Adaptive parameter ownership in the compact population interface. The seven neuron/class coefficient groups are detailed in Appendix A.5.

Group	Ownership	Coefficients	Effect
Behavior	one neuron	88 per neuron	input, memory and readout
Morphology	one structural class	88 per class	class-shared response
Synaptic	one contact	one per contact	positive input efficacy

4.2 Full temporal gradients and captured updates

Full backpropagation through time differentiates the response to the complete supplied event history, from contact efficacy and branch aggregation through every recurrent state transition to the sequence loss. Behavior, morphology and synaptic banks can be optimized separately or together. This supplied-history trajectory is the object of temporal differentiation; autonomous connected inference instead forms discrete events by thresholding the predicted logits.

Sparse event indices and values are accumulated into branch features before temporal integration, avoiding a dense neuron–time–contact history. When activation storage is large, neuron chunks retain the complete temporal horizon and share the class bank; their neuron-count-weighted losses contribute to one optimizer update. Gradient checks compare the behavior, morphology and synaptic derivatives with the unchunked tensor computation.

The accelerated update captures input processing, forward execution, loss, backward propagation, clipping and optimization in one CUDA Graph, using PyTorch’s fixed-shape, fixed-address replay mechanism [16]. Each replay copies fresh inputs and targets into the captured buffers. Warmup and capture restore both model and optimizer state before use, so preparing the graph preserves the caller’s initial parameters and update count.

A matched 40-update SHD comparison produced zero maximum absolute parameter discrepancy between eager and captured execution. Eager updates took 6.40 s and replay took 1.84 s, a 3.49-fold

speedup, with 0.74 s of separate capture preparation. The same source state, ordered batches, Adam optimizer, cross-entropy loss and clipping rule were used in both paths (Appendix A.8). Dense-contact, real-time full-BPTT measurements are reported separately in Section 9.

5 Evaluation methodology

5.1 Separating fitting, selection, and evaluation

Each comparison separates data used to fit parameters, choose a model or adaptation schedule, and estimate final performance. The unit of separation follows the scientific question. Ordinary AxoBench uses morphology-balanced traces, population specialization uses recurrent contexts, SHD uses disjoint examples and independently trained source models, and connected evaluation uses unseen connectomes. Appendix A.1 records the exact partitions.

For the ordinary comparison, development data fix the model identity and threshold-calibration rule. The first 24 final traces determine one prevalence-matched threshold, and the remaining 96 provide all reported fidelity metrics. Released Branch-ELM models are evaluated unchanged.

Population-input specialization selects one learning-rate schedule per architecture family on development contexts and applies it unchanged on the final contexts. SHD selects the adaptive groups, learning rate, and update count on one source model, then reuses that choice for five independently trained sources. Connected-population comparisons pair alternatives on each unseen connectome. Final data are therefore used only after model and schedule choices have been fixed.

5.2 Neuron and population fidelity

The single-neuron evaluation follows the AxoBench core metric set [25]. Spike Mean F1 averages event F1 over integer matching tolerances from 0 through 5 ms,

$$\text{MeanF1}_{0:5} = \frac{1}{6} \sum_{\tau=0}^5 F_1(\tau).$$

Predicted and teacher events are matched one-to-one in temporal order within $\pm\tau$. The six constituent values distinguish exact event recovery from agreement that appears only at a permissive tolerance.

For prediction error $e_i = \hat{y}_i - y_i$ and teacher-derived relevance $r_i \in [0, 1]$, the conditional squared error above threshold u is

$$E(u) = \frac{1}{|\{i : r_i \geq u\}|} \sum_{i:r_i \geq u} e_i^2.$$

SERA is the support-normalized integral of $E(u)$, evaluated at 101 evenly spaced relevance thresholds. Voltage SERA applies the construction to somatic voltage and has units of mV^2 . Dynamics SERA applies it to adjacent native-time voltage changes and has units of mV^2/ms^2 . SERA already names the squared quantity, and Root-SERA denotes its square root. Lower SERA is better.

Response SERA evaluates paired changes in voltage after an intervention. It compares the predicted intervention-minus-baseline change with the teacher’s change, using relevance derived from both teacher trajectories and their difference. This tests the accuracy of the response to altered input; its units are mV^2 .

Connected evaluation keeps three discrepancies separate. A morphology term compares morphology-conditioned firing rates with fixed targets, a temporal term compares input-count autocorrelations at lags from 1 to 256 ms, and a stationarity term measures full-to-tail drift in global and role-conditional

activity. Global rate, input load, variability, and active fraction are also checked for collapse or composition changes. The equations and fixed normalizations appear in Appendix A.16.

5.3 Uncertainty and runtime contracts

Single-neuron differences use 1,000 paired bootstrap resamples of the 96 scored traces, stratified by morphology. All three metrics are recomputed from each paired sample. These intervals describe trace-level uncertainty for fixed trained models. Intervention comparisons instead resample 175 paired shard clusters within 25 morphology-condition strata and use simultaneous intervals for all twenty AxoSim-Branch-ELM response differences. Retraining variation requires independently fitted models, as used for adaptation, where the report gives the mean and sample standard deviation across source models. Population-training results use the paired outcome on every unseen connectome. The reported uncertainty therefore follows the source of variation in each analysis.

Runtime quantities are attached to explicit workloads. Isolated-neuron rates use float32 and matched batch and sequence shapes. Population inference uses the p90 latency of a complete step, including event discovery, delayed routing, branch accumulation, mutable efficacy, recurrent state, and 1% telemetry. Population adaptation measures a complete 256-step update, including transfer, sparse input construction, forward execution, loss, full-sequence backward propagation, clipping, and optimization. Hardware, precision, repeats, and exclusions are tabulated in Appendix A.2.

CoreNEURON, Branch-ELM, and AxoSim use different workloads and hardware, so Figure 1 compares their directly measured quantities. A cross-system speed ratio is reserved for a future shared-workload measurement. Raw values and axis mappings appear in Appendix A.11.

6 Single-neuron fidelity and efficiency

6.1 Held-out fidelity and the practical model frontier

The largest quantitative separation in the ordinary AxoBench results occurs in spike timing (Table 3). AxoSim-GRU Medium reaches Mean F1 0.522, whereas the strongest released Branch-ELM spike reference reaches 0.077, an absolute difference of 0.445. At the same time, GRU Medium lowers Voltage SERA from 307.18 to 241.61 mV², a 21.3% reduction. Mamba Small lowers Dynamics SERA from 448.57 to 428.72 mV²/ms², a smaller 4.4% reduction. The dominant improvement is therefore event recovery, accompanied by a substantial voltage-trajectory gain and a modest improvement in native-time voltage changes.

Table 3: Held-out ordinary-AxoBench results. Lower SERA and higher Mean F1 and throughput are better. Inference is measured in millions of neuron-steps/s.

Model	Parameters	Voltage SERA	Dynamics SERA	Mean F1	Inference
AxoSim-GRU Small	17,432	243.92	429.12	0.486	21.79
AxoSim-GRU Medium	122,744	241.61	429.13	0.522	16.81
AxoSim-Mamba Small	32,536	244.29	428.72	0.454	18.86
AxoSim-Mamba Medium	1,128,200	244.46	439.00	0.520	4.40
Branch-ELM M30	10,952	307.41	448.57	0.077	0.742
Branch-ELM M50	19,252	307.18	448.58	0.076	0.735

For AxoSim-Mamba Medium, Mean F1 is 0.443 higher than the strongest Branch-ELM spike reference, with a paired morphology-stratified 95% bootstrap interval of [0.422, 0.463]. Its Voltage-

SERA difference from the strongest released voltage reference is -62.72 mV^2 , with a 95% interval of $[-69.06, -56.72]$. AxoSim-GRU Medium has the highest final Mean-F1 point estimate, 0.0016 above Mamba Medium. The near tie in spike fidelity is practically important because the GRU model is much smaller and nearly four times faster at inference.

The target quantity determines model choice (Figure 4). GRU Medium leads Mean F1 and Voltage SERA, GRU Small leads inference throughput, and Mamba Small records the lowest Dynamics SERA. Mamba Medium provides a state-space alternative within 0.002 of the spike leader, at 4.40 million neuron-steps/s. The three leaders define the measured frontier. The larger Mamba state offers temporal capacity at a higher execution cost.

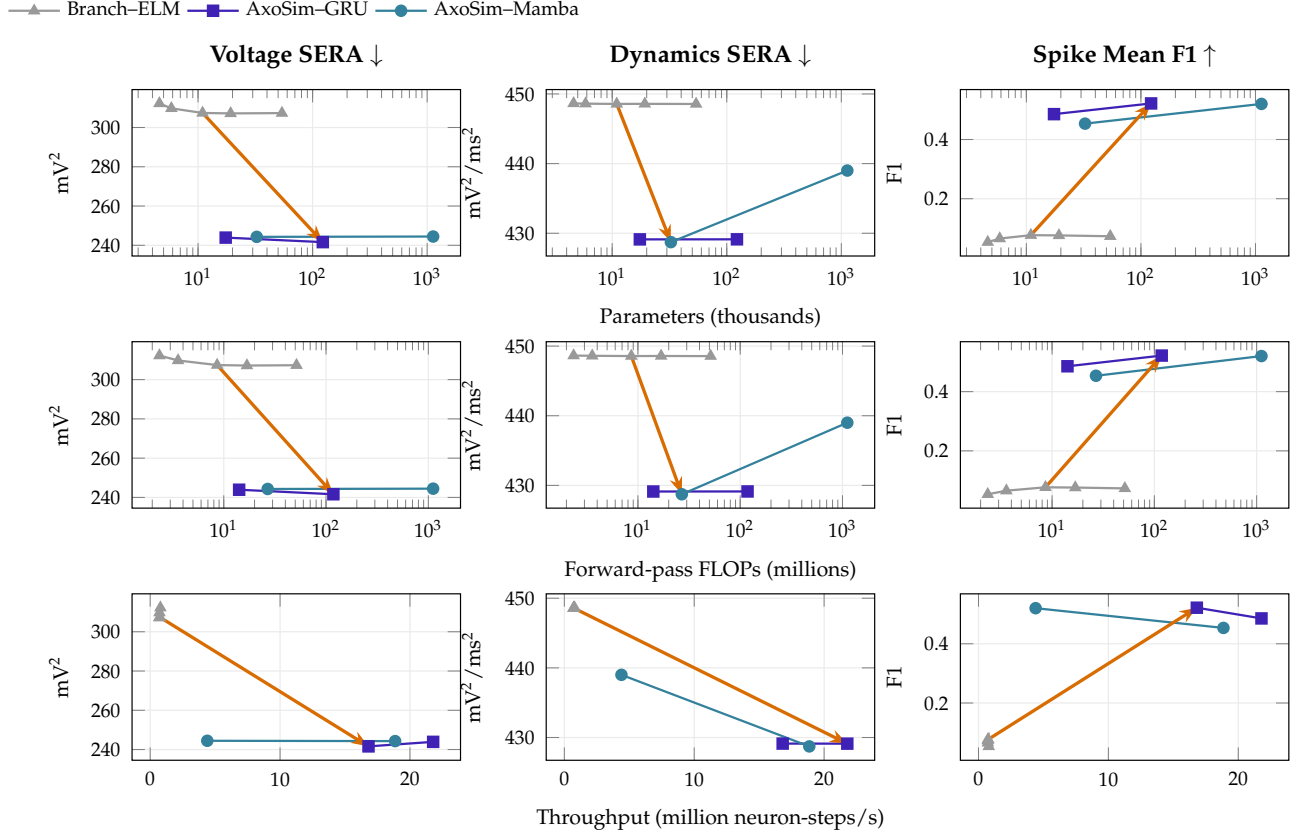


Figure 4: Accuracy–cost tradeoffs for AxoSim and released Branch-ELM models. Voltage SERA, Dynamics SERA and spike Mean F1 are shown against trainable parameters, forward-pass FLOPs and inference throughput. Lines connect models from the same architecture family, and orange arrows link the Pareto-selected Branch-ELM and AxoSim models. FLOPs are the operations recognized by `torch.profiler` for a batch-one, 500-step forward pass with 1,278 input channels. Throughput uses an RTX 5090, batch 32, 500 steps and float32. The Pareto rule appears in Appendix A.12.

6.2 Where the fidelity gains arise

Mean F1 averages six spike-matching tolerances, and AxoSim’s advantage appears at every setting. GRU Medium exceeds Branch-ELM M30 at every tolerance in Figure 5. At exact timing, F1 is 0.261 versus 0.047, and at 5 ms it is 0.641 versus 0.102. The throughput-leading GRU Small has the same qualitative separation from the closest-size Branch-ELM reference.

Conditional squared-error curves provide the complementary voltage result. Integration of each conditional squared-error curve yields SERA. Across most of the relevance support, AxoSim has lower conditional voltage error than Branch-ELM, so the Voltage-SERA reduction is distributed across a broad threshold range. The smaller separation between the dynamics curves agrees with the modest aggregate Dynamics-SERA improvement. Thus, event recovery can improve markedly even when error in native-time voltage changes decreases only slightly.

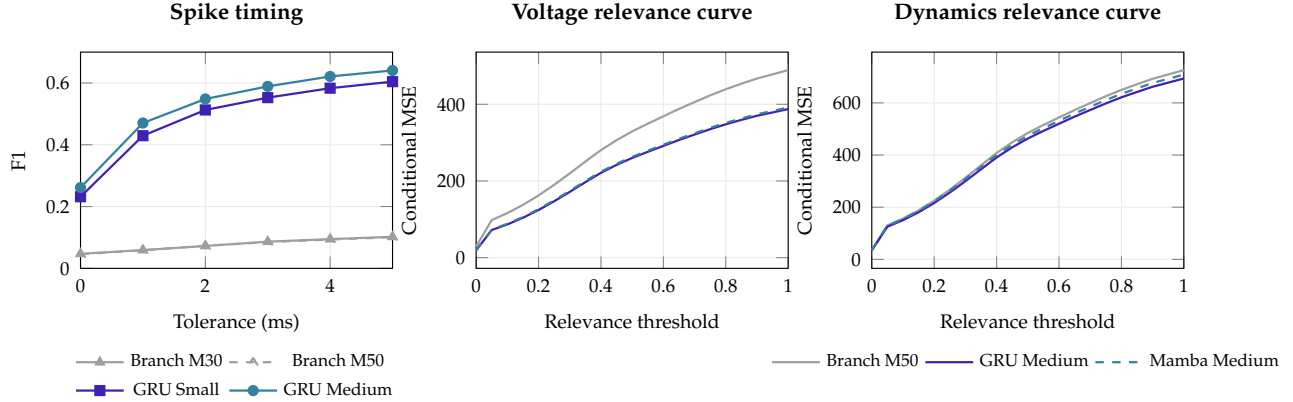


Figure 5: Constituents of the core fidelity metrics. Spike F1 is reported across matching tolerances, followed by the conditional squared-error curves integrated by Voltage and Dynamics SERA.

7 Intervention-response fidelity

7.1 Predicting the teacher’s response to changed synaptic input

Controlled changes in synaptic input provide a direct test of response prediction. For each baseline–intervention pair, Response SERA compares the predicted voltage change with the detailed teacher’s change using the AxoBench relevance weighting [25]. Four AxoSim models and five released Branch-ELM sizes are evaluated on the same 350 paired histories spanning event removal, excitatory removal, inhibitory removal, site silencing and temporal jitter across five morphologies. Model weights remain fixed, and 50 separate baseline pairs determine the spike thresholds before scoring (Appendix A.9).

AxoSim reaches Response SERA $255.55\text{--}258.78\text{ mV}^2$, compared with 286.17 mV^2 for the strongest Branch-ELM response model. The resulting reductions are 9.57–10.70%, and all twenty AxoSim–Branch-ELM paired comparisons have simultaneous 95% intervals below zero. A zero-voltage-change prediction gives 298.27 mV^2 , providing a reference for the amount of response captured. Spike Mean F1 on the intervened traces is 0.416–0.491 for AxoSim, versus 0.062–0.076 for Branch-ELM.

The condition-wise firing changes make these responses concrete (Figure 6). Removing inhibitory events increases the teacher’s mean firing rate by 4.36 Hz; GRU Medium and Mamba Medium predict increases of 3.09 and 3.45 Hz, while Branch-ELM M1 predicts 17.98 Hz. Conversely, removing excitatory events decreases teacher activity by 3.47 Hz, compared with 3.36 and 3.47 Hz for the two AxoSim models. Event removal and site silencing likewise produce teacher-aligned decreases. These observations relate the output-fidelity comparison to the expected effects of changing excitation and inhibition.

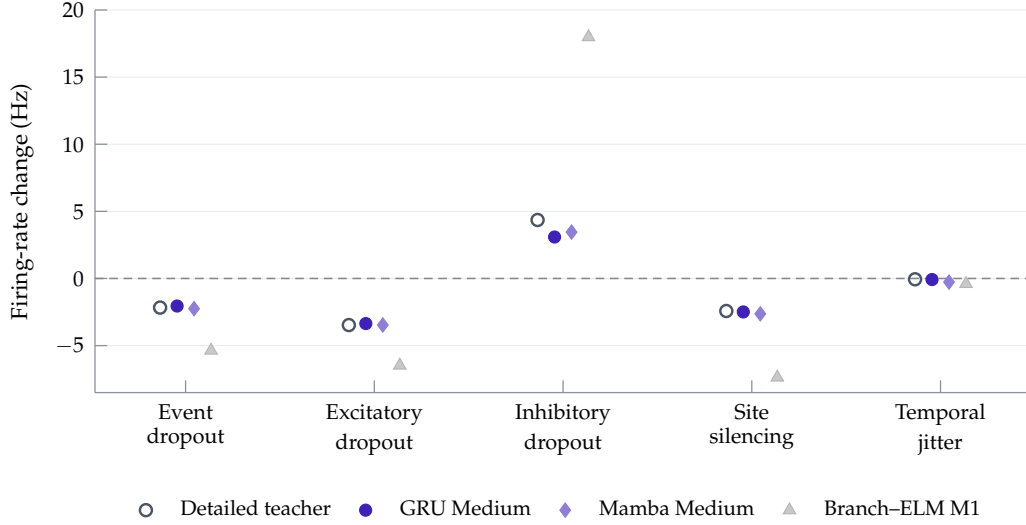


Figure 6: Mean firing-rate changes under five synaptic-input interventions. Points show the detailed teacher and three models on the same 350 paired scoring histories, excluding the first 500 ms of each six-second history. Thresholds are fixed from separate baseline calibration pairs. Branch-ELM M1 has the lowest pooled Response SERA among the released Branch-ELM models. Full response metrics and sampling are given in Appendix A.9.

Predicted voltage changes also correlate more closely with the teacher, with centered Pearson correlations of 0.483–0.487 for AxoSim versus 0.331–0.336 for Branch-ELM. At the same time, the least-squares response gain through zero remains below one, at 0.269–0.303 for AxoSim and 0.179–0.186 for Branch-ELM, where a gain of one denotes the teacher’s voltage-change scale. Correlation and gain therefore describe complementary aspects of the lower Response SERA.

7.2 Internal-state diagnostics of intervention responses

Mamba Medium’s internal states provide a complementary description of the predicted responses. A separate analysis uses all 12,500 intervention pairs to form 1,575 disjoint cohort signatures, each averaging the paired temporal state displacement of seven or eight traces over successive 500-ms windows. The three-dimensional UMAP projection shows intervention-dependent regions while marker shape identifies morphology (Figure 7a; 12). Quantitative comparisons use the original representation space.

Temporal trajectories use a separate, deterministic selection of 200 pairs. The first three principal components explain 83.2% of their standardized state-displacement variance, and condition means trace the evolution of the response at 500-ms intervals (Figure 7b). Gray arrows summarize average next-window displacement within occupied spatial bins, following low-dimensional analyses of recurrent dynamics [4, 21]. Across the intervention–morphology groups, within-intervention directions exceed between-intervention directions by 0.563 in cosine similarity, describing consistent response organization across morphologies.

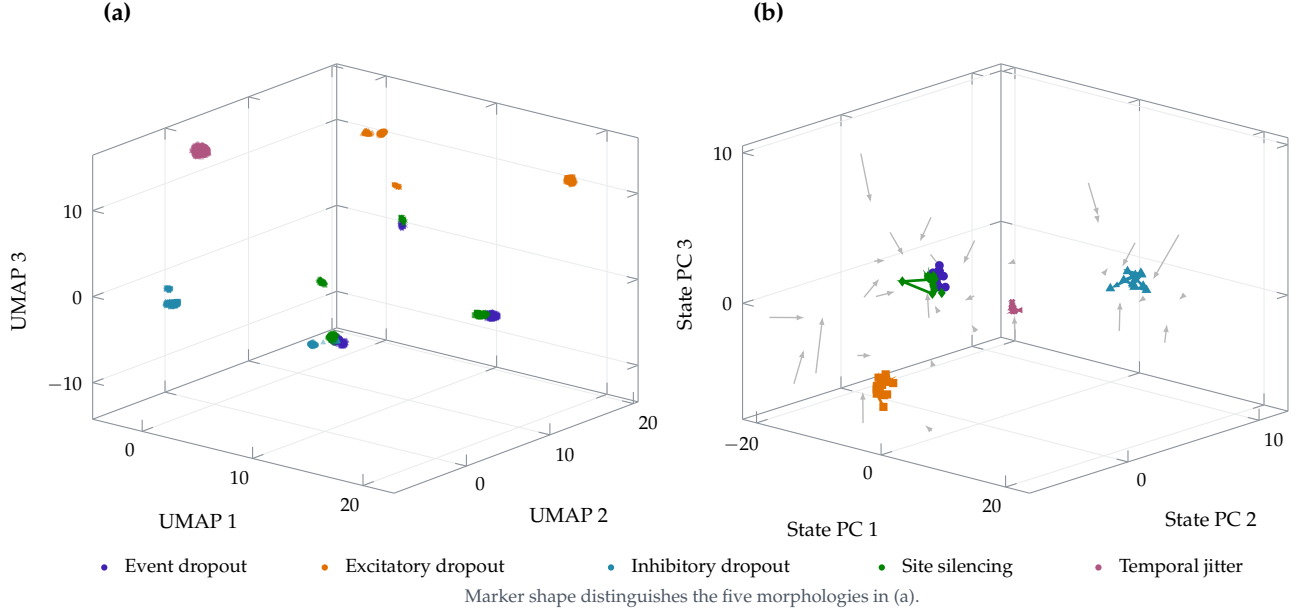


Figure 7: State-response diagnostics for Mamba Medium. (a) Three-dimensional UMAP of 1,575 disjoint cohort-averaged signatures from 12,500 pairs. Color denotes intervention and marker shape denotes morphology. (b) Average temporal responses in three PCA coordinates, with arrows showing mean movement within occupied spatial bins. Paths use 200 deterministically selected pairs; the first three components explain 83.2% of standardized variance. Output-response metrics are reported separately in Appendix A.9.

An external response distance links this organization to teacher outputs, following scientific-model manifold comparisons [15] and representational similarity analysis [10]. Across all 300 pairs of intervention–morphology groups, original-space temporal-state geodesics correlate 0.446 with the teacher’s voltage-and-spike response distance. The corresponding branch representation reaches 0.211, and the input-perturbation descriptor reaches 0.471. Temporal-state distances therefore associate more closely with teacher responses than branch distances, while input descriptors show a comparable association. The distance construction, paired uncertainty and all three comparisons appear in Appendix A.10.

8 Connected population simulation

8.1 Throughput as a function of connectivity

The population benchmark advances 1,048,576 heterogeneous neurons at native 1 ms resolution. One real-time second therefore requires 1.048576×10^9 neuron-steps. The measured step includes event-derived firing, delayed contact routing, branch accumulation, mutable efficacies, recurrent state, and pinned-host telemetry for 1% of the neurons.

Throughput decreases as incoming synaptic connectivity increases (Figure 8). The million-neuron measurement uses 362 incoming contacts per neuron and advances at 1.177×10^9 neuron-steps/s. At 4,000 contacts, a closed-loop 65,536-neuron population advances at a median 2.215×10^8 neuron-steps/s, equivalent to 3.38 times real time. A routing-density sweep at the same population spans 100 to 10,000 contacts while holding source activity, external input width, and telemetry constant. Its 10,000-contact endpoint remains 2.17 times real time.

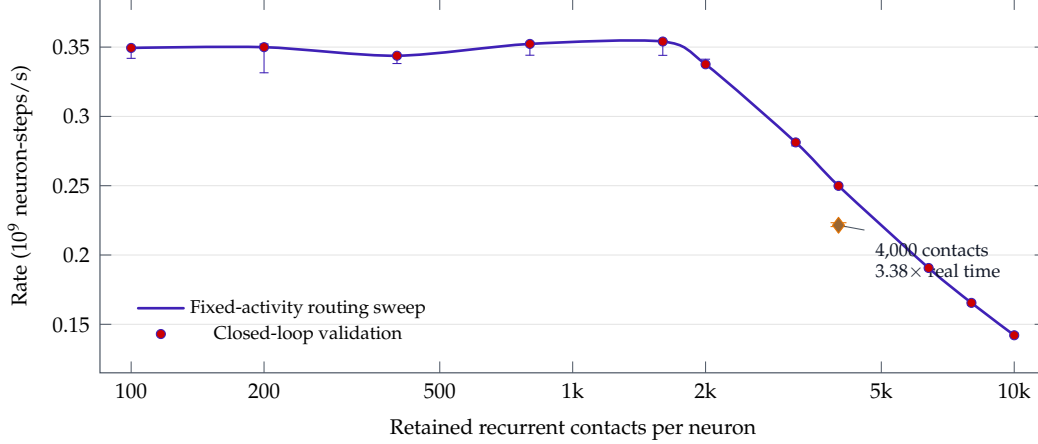


Figure 8: Connectivity scaling for a fixed population of 65,536 neurons. Each marker is the median of three p90 rates, and vertical ranges span three independently realized graphs. The line connects measurements with source events replayed at morphology- and role-conditioned target rates, isolating routing density from changes in activity. The orange diamond validates the 4,000-contact point with threshold-derived closed-loop firing. Every timed step includes the threshold scan, delayed routing, branch accumulation, an independently mutable efficacy for every contact, recurrent state, and telemetry for 1% of neurons; external afferent width remains fixed.

Population scale, connectivity, and simulated speed form a three-way tradeoff. The million-neuron measurement reports connectivity at a fixed population scale. The 65,536-neuron measurements extend the evidence to thousands of recurrent contacts and separate closed-loop execution from routing-density scaling. Appendix A.13 defines both workloads and reports the complete sweep.

8.2 AxoBench traces from recurrent population contexts

Recurrent feedback produces a distinct input distribution because event timing, excitation and inhibition, morphology, and graph structure jointly change branch load and temporal correlation. To sample that distribution, a thresholded AxoSim network generates branch-resolved contact histories, which are then replayed through matching detailed teachers. The recurrent learned system supplies the inputs, and five excitatory morphologies together with one inhibitory morphology supply detailed voltage and spike labels.

The axobench-population release contains 50,000 native-millisecond traces from 100 independently generated contexts. Eighty contexts provide 40,000 training traces, and 20 disjoint contexts provide 10,000 validation traces. Both splits preserve an 80/20 excitatory–inhibitory ratio. Validation recomputes split membership and checks manifest hashes, identifiers, array shapes, finite values, labels, and generator provenance. Paired perturbations reside in the separate axobench-interventions dataset, preserving the base population distribution.

Table 4: Composition of axobench-population. Complete recurrent contexts define the train-validation split.

Split and role	Traces	Contexts	Mean firing rate (Hz)
Train, excitatory	32,000	80	5.62
Train, inhibitory	8,000	80	6.32
Validation, excitatory	8,000	20	5.71
Validation, inhibitory	2,000	20	6.29
Total	50,000	100	–

8.3 Trace fidelity and recurrent validity are distinct

A plausible global firing rate can coexist with incorrect morphology-conditioned rates, temporal correlations, or late activity drift. A paired comparison isolates the effect of recurrent context by training one model on population-context examples and another on the original single-neuron distribution. On three unseen 4,000-neuron connectomes, population-context training reduces morphology discrepancy by 51.1%, temporal discrepancy by 7.2%, and stationarity discrepancy by 55.1%. It satisfies all three criteria fixed before evaluation, while the ordinary-input baseline misses the morphology criterion.

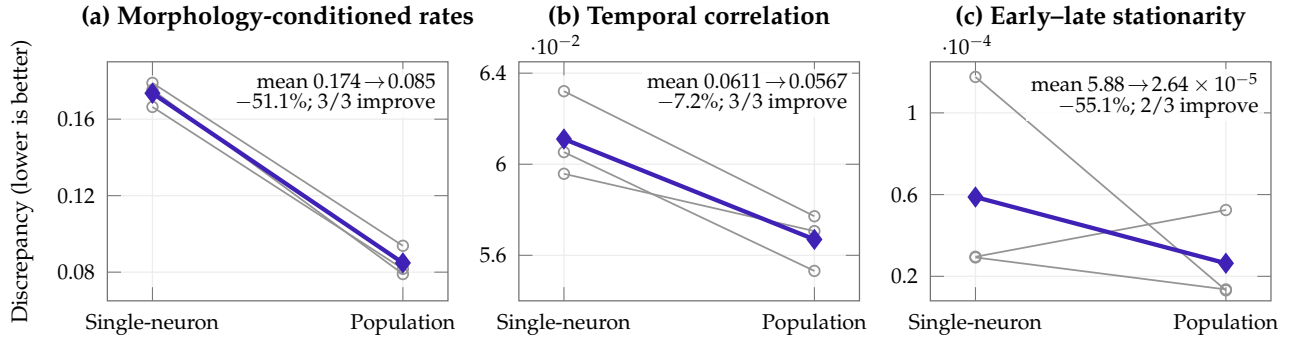


Figure 9: Development-corpus population-context training on three unseen 4,000-neuron connectomes. Thin lines pair training distributions on the same connectome; the purple diamond line connects cross-connectome means. Morphology measures mean squared log error in morphology-conditioned firing rates, temporal discrepancy measures lag-correlation error, and stationarity measures early-late log-ratio error. Annotations report the mean change and the number of improving connectomes. Lower is better throughout.

Training on all 50,000 traces improves held-out Voltage SERA from 299.99 to 285.84 mV², Dynamics SERA from 501.67 to 483.41 mV²/ms², exact-spike F1 from 0.136 to 0.163, and Mean F1 from 0.331 to 0.340. Yet the connected model misses the morphology-conditioned criterion on all three unseen connectomes. Consequently, improved trace-level metrics do not imply recurrent validity. Connected measurements determine whether the gain transfers to recurrent deployment.

9 Inference-time adaptation

9.1 Structural and synaptic transfer

Five leave-one-morphology-out comparisons measure specialization of a small structural adapter to an excluded morphology. Each source uses the other four morphologies, whereas target adaptation and evaluation use disjoint stimulus schedules from the omitted morphology. Updating one 58-value row improves Mean F1 and Dynamics SERA on all five targets. Voltage SERA improves on three, while the other two change by only +0.27% and +0.37%. Full tuning of 38,257 parameters provides an upper reference for the attainable gain.

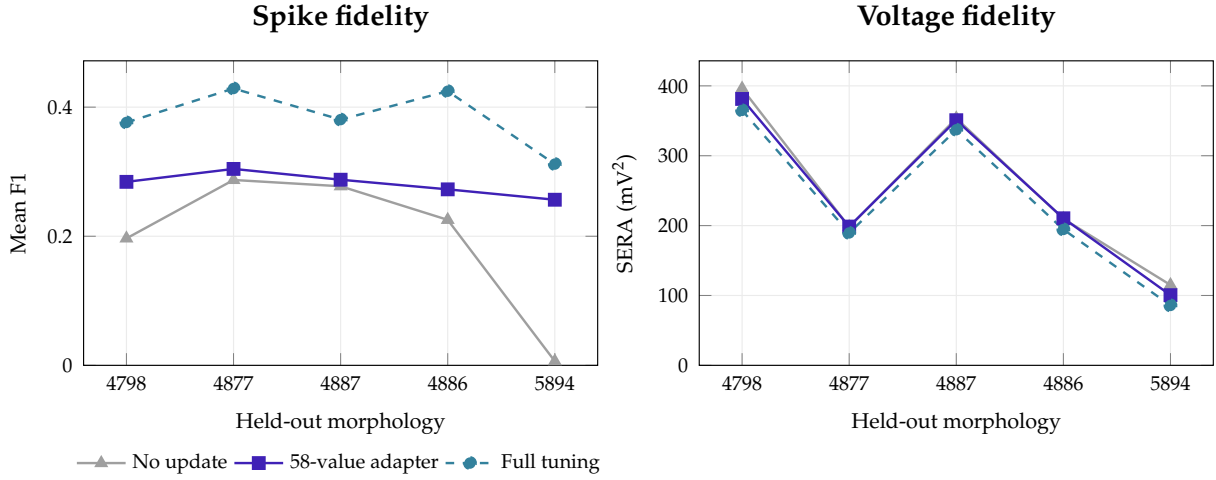


Figure 10: Adaptation to five separately excluded morphologies. Each line follows one adaptation alternative across the five held-out targets. The 58-value adapter improves spike and dynamics fidelity for every morphology; full tuning provides an upper reference.

Where morphology transfer targets an omitted class, synaptic transfer addresses a known perturbation. Two conditions of one Hay layer-5 pyramidal-cell teacher differ only by an excitatory-conductance scale of 0.925 versus 1.0, with all 1,278 physical input channels. In the 0.925-to-1.0 direction, contact adaptation reduces Voltage SERA by 4.76%, reduces Dynamics SERA by 0.31%, and raises Mean F1 by 0.044. The learned contact changes correlate 0.724 with the known conductance intervention. In the reverse direction, correlation rises to 0.834 but Voltage SERA worsens. In the forward direction, contact coordinates both track and predictively compensate for the controlled shift. The reverse direction demonstrates the insufficiency of parameter correlation alone for a biological interpretation.

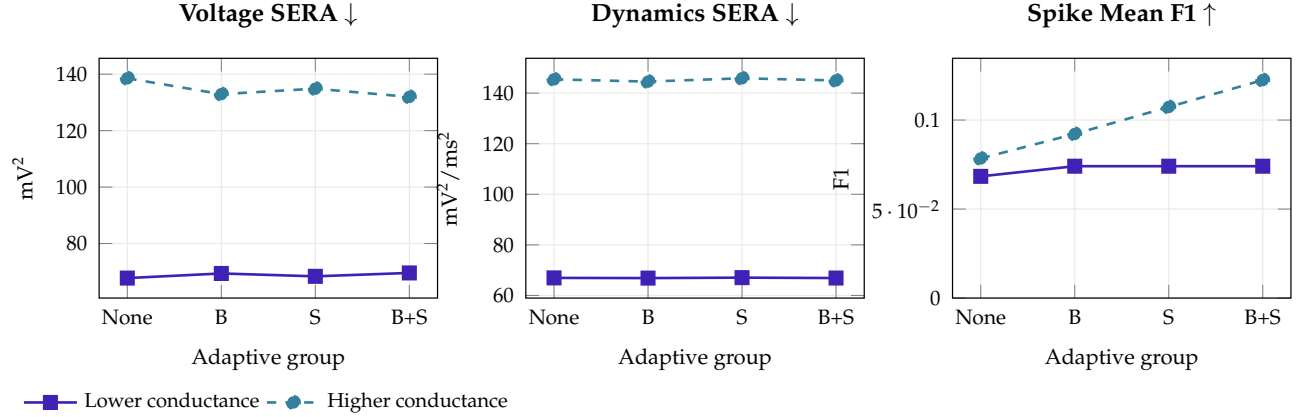


Figure 11: Transfer between matched conductance conditions of one detailed teacher. The 0.925-to-1.0 direction improves all three fidelity metrics. The reverse direction follows the imposed channel pattern more closely while increasing held-out Voltage SERA.

The role of each adaptive coordinate becomes concrete in these transfer comparisons. Updating a morphology row applies one class-shared correction, whereas synaptic updates change individual contact efficacies. Predictive improvement after a controlled shift demonstrates their practical utility. Cellular identification requires consistent predictive success across interventions.

9.2 Specialization to recurrent input distributions

Population-context inputs differ from the ordinary AxoBench distribution even when the neuron target is unchanged. Four AxoSim models and the released Branch-ELM family are adapted for 30,000 recurrent-domain presentations using the same context partition and architecture-specific schedules fixed on development data. Performance is then measured on 20 untouched target contexts.

Every AxoSim model improves target Mean F1. Mamba Medium reaches 0.412 after adaptation, compared with 0.136 for the strongest Branch-ELM target result. The other AxoSim models reach 0.364–0.402 and preserve the same large event-recovery separation. Target-domain results appear in Table 5; the full metric set is given in Appendix A.14.

Table 5: Spike fidelity after specialization to recurrent population inputs. The improvement is relative to the same model before target-domain adaptation.

Model	Target Mean F1	Improvement
AxoSim-GRU Small	0.379	+0.066
AxoSim-GRU Medium	0.402	+0.073
AxoSim-Mamba Small	0.364	+0.064
AxoSim-Mamba Medium	0.412	+0.094
Branch-ELM M100	0.136	+0.055

Corpus training targets a generally deployable recurrent model and therefore requires connected validation. Recurrent-domain adaptation instead measures whether exposed coordinates improve behavior in a specified target domain. The same data source supports both analyses under these distinct evaluation contracts.

9.3 Adaptation across a temporal task shift

Spiking Heidelberg Digits (SHD) supplies an external temporal distribution shift [5]. Development data compare seven optimizer–learning-rate–duration schedules and every nonempty combination of behavior, morphology, and synaptic groups. Morphology-plus-synapse adaptation, a learning rate of 2×10^{-3} , and 40 updates are fixed on one source model and reused for five independently trained sources. Each run uses 160 adaptation examples, 80 development examples, and 800 final examples, all disjoint.

Across six independently trained source models, target accuracy rises from 64.50% to 70.96%. The mean gain is 6.46 ± 1.20 percentage points, where the uncertainty is the sample standard deviation across six source models, and every target comparison improves. The source classifier used to choose the adaptation schedule reaches 91.88% before the temporal shift, localizing the lower target value to the shifted task.

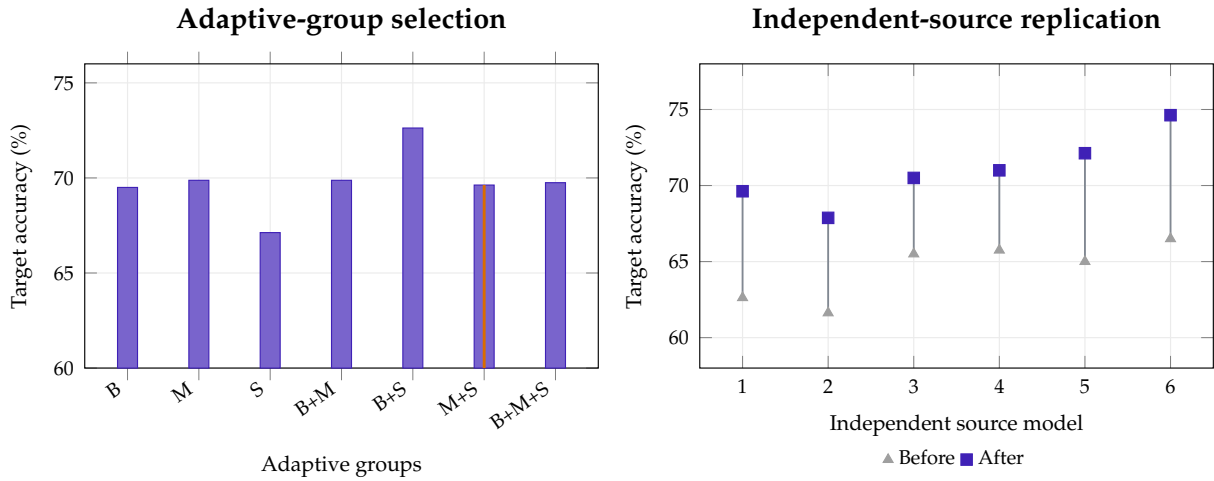


Figure 12: SHD accuracy before and after inference-time adaptation. One source model fixes the adaptive parameter groups and update schedule; five independently trained models reuse it. Target accuracy improves in all six comparisons.

Accuracy gains are non-monotonic in contact count. At 700 contacts, two independently trained source models improve from 65.13% to 72.13%, a gain of 7.00 ± 1.06 points, while their mean source validation accuracy is 92.56%. The complete contact-density analysis appears in Appendix A.15. The external task therefore provides replicated evidence of useful within-system adaptation. Comparison with separate test-time-adaptation algorithms remains future work.

9.4 Population size, connectivity, and update time

The runtime contract measures one complete 256-step adaptation update. Each replay includes fresh pinned-host input transfer, exact sparse contact aggregation, forward execution, loss, full backward propagation, gradient clipping, and one capturable-Adam update in BF16. Real-time adaptation requires all seven changing replays to finish within 256 ms.

At 2,000 retained contacts per neuron, all seven updates remain below the deadline for 20,000 neurons, with a slowest update of 252.17 ms. The next measured population, 21,000 neurons, exceeds the deadline. Doubling connectivity to 4,000 contacts reduces the real-time population to 10,000 neurons,

whose slowest update takes 250.50 ms. Behavior, morphology, and synaptic parameters are optimized together, and every retained contact has an independently trainable efficacy.

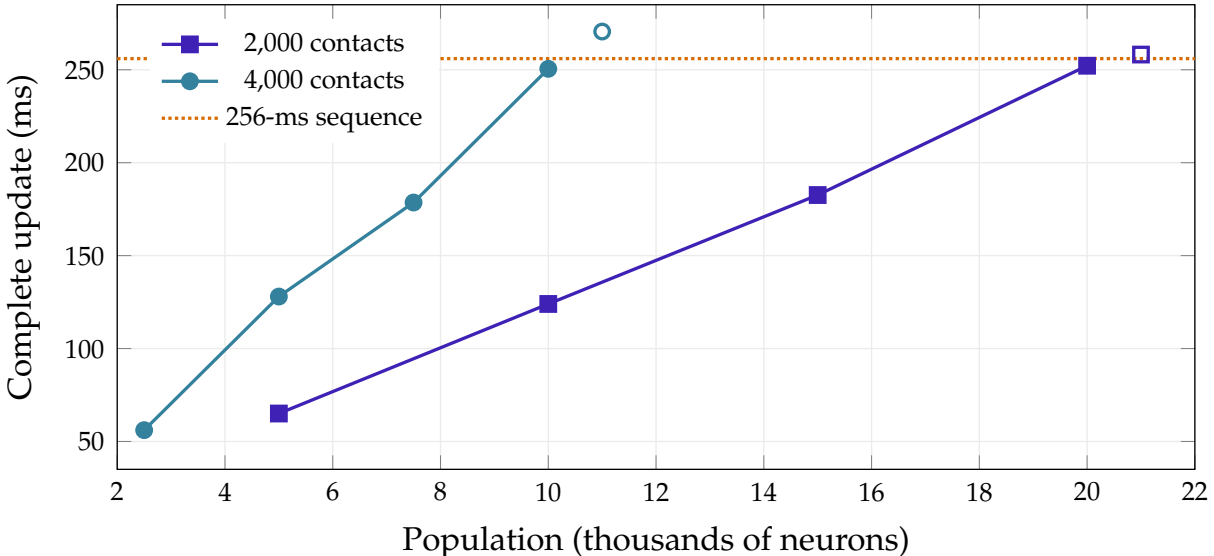


Figure 13: Complete full-BPTT update time for densely connected populations. Markers show the slowest of seven changing updates at each population size; filled markers below the dotted 256-ms boundary satisfy the real-time criterion, while open markers show the first measured misses. Each update includes fresh pinned-host input, exact sparse aggregation, forward execution, loss, full backward propagation, gradient clipping, and one Adam step with all adaptive parameter groups enabled.

Repeated fixed-shape updates therefore support full-sequence learning at the reported population and connectivity. Section 11 discusses sustained long-horizon online learning as the next research direction.

10 Relation to learned and mechanistic simulators

10.1 Learned and mechanistic neuron models

Temporal neural networks can approximate the millisecond input–output map of a morphologically detailed cortical neuron and amortize repeated numerical integration of the teacher [2]. Spieler et al. introduced the Expressive Leaky Memory neuron as a compact alternative in which dendritic integration and slow memory reproduce detailed-neuron responses with relatively few parameters [19]. AxoSim uses released Branch-ELM models as its principal learned reference and the Hay layer-5 pyramidal-cell family as a principal mechanistic teacher [8].

Arbor and CoreNEURON instead accelerate networks of morphologically detailed neurons while retaining compartments, membrane mechanisms, and their mechanistic interpretation [1, 11]. Jaxley makes detailed simulation differentiable with respect to explicit morphological, synaptic, and ion-channel parameters [6]. These systems remain appropriate for studying the effect of a named conductance or compartment. AxoSim amortizes detailed teacher simulations by fitting each learned model once to generated responses and then reusing it for population realizations and gradient-based specialization within the learned domain.

10.2 Simulation abstractions and population execution

Brian 2 separates model equations from generated execution code, allowing a scientific description to target efficient backends [20]. BrainPy similarly combines dynamical systems, transformations, simulation, training, and analysis in one programming model [24]. The same separation is important for learned neurons. AxoSim exposes ordinary PyTorch modules for model construction and differentiation, then maps the compact population model to fused kernels while preserving contact identity, delay semantics, mutable efficacy, and parameter ownership.

Population execution also changes what must be measured. Cortical synapse counts vary by species, region, layer, and cell type, with relevant values in the thousands across reported preparations [18]. Accordingly, AxoSim attaches each population throughput measurement to a specific number of incoming synaptic contacts. Connected validation then measures morphology-conditioned responses, temporal correlations, and stationarity as a distinct level of recurrent evidence.

10.3 Metrics for learned neuron simulation

AxoBench introduced the core evaluation set used here, consisting of tolerance-dependent spike F1, Voltage SERA, and Dynamics SERA [25]. SERA integrates conditional squared error over teacher relevance and follows the imbalanced-regression construction of Ribeiro and Moniz [17]. Spike F1 is reported across explicit timing tolerances, complementing established spike-train distances [22, 23].

Each quantity measures a different aspect of the response. Spike F1 measures event recovery, Voltage SERA measures errors in relevant parts of the somatic trajectory, and Dynamics SERA measures errors in native-time voltage changes. A model can therefore receive different rankings on the three metrics. For recurrent use, morphology-conditioned activity, temporal correlation, and stationarity diagnostics extend the metric set to population scale.

11 Discussion and future work

11.1 Recurrent use changes the unit of validation

Learned simulation creates a hierarchy of validation levels. Held-out traces measure voltage and spike prediction under externally specified input. Connected evaluation then measures behavior after those predictions alter the next input distribution, while adaptation differentiates the temporal response to a complete supplied history. Each level therefore requires its own evidence.

The distinction is visible in the population-data comparison. A controlled context corpus improves morphology, temporal, and stationarity behavior on unseen connectomes. Training on all 50,000 traces improves every held-out trace metric under the fixed training schedule, yet the connected model misses the morphology criterion. Small conditional errors can move recurrent state into a different input regime, and the graph then determines how those errors propagate. Consequently, a learned simulator should be validated at the level where its predictions become interventions.

These observations motivate an iterative division of labor. Detailed models generate teacher responses and controlled perturbations. Learned models amortize those responses across many connectomes or optimization steps. Connected diagnostics then identify distribution shifts and trigger new teacher simulations before the next deployment regime.

11.2 Edge-associated state dominates population scale

Parameter count of the shared neuron model is an incomplete predictor of population cost. At one million neurons, routing and branch accumulation grow from 59.7% to 78.8% of the component-median sum as incoming connectivity increases from 100 to 362 contacts. The fixed-activity routing sweep at 65,536 neurons then shows a broad plateau through 1,600 contacts, followed by a decline from 0.338 billion neuron-steps/s at 2,000 contacts to 0.142 billion at 10,000. Inference rate therefore depends jointly on population size, connectivity, activity, telemetry, and hardware.

Adaptation makes this relationship sharper because every retained edge also owns differentiable state. Doubling the retained contacts from 2,000 to 4,000 halves the measured real-time full-BPTT population from 20,000 to 10,000 neurons. Compact shared weights can be reused across the population, whereas edge-associated routing, storage, and gradients scale with the realized graph. Reports of learned population simulators should therefore attach neuron count to contact count and update mode.

11.3 Learned coordinates require predictive interventions

Explicit parameter ownership permits optimization at neuron, structural-class and contact scales. Morphology rows improve transfer to five omitted structures, and synaptic values can follow a known conductance pattern. The class and neuron coefficients enter additively, so biological attribution requires predictive validation beyond ownership. In the reverse conductance transfer, parameter correlation is higher even though Voltage SERA worsens. A learned coordinate gains biological relevance when its change predicts the response to a controlled intervention.

The matched intervention comparison supplies direct output evidence through lower teacher-response error and teacher-aligned firing changes. Internal state geometry complements this predictive test. Its association with teacher-response distance rises from 0.211 in the branch representation to 0.446 after temporal integration, while the input perturbation reaches 0.471. The geometry consequently describes both the imposed input structure and its transformation through temporal memory. Together, output prediction, external response distances and controlled transfer relate learned coordinates to observable neuronal responses.

11.4 Future work

Population-aware hyperparameter and objective selection on disjoint development connectomes should precede one evaluation on untouched connectomes. Such a comparison would determine whether trace-level gains transfer to recurrent behavior and whether mixed-domain training preserves broad single-neuron fidelity.

The AxoBench metrics are also unsaturated. Replicated training should examine depth, width, residual conditioning, objective balance, and compute while improving spike Mean F1, Voltage SERA, and Dynamics SERA together. In the measured frontier, the largest temporal core approaches the spike leader while smaller models lead the remaining metrics. Controlled, replicated trajectories can determine how depth, width, and compute produce that tradeoff.

Adaptation should extend from bounded transfer episodes to sustained long-horizon online learning. Such a study must measure cumulative task performance, memory growth, numerical stability, and drift over many updates. A matched-simulator comparison should fix morphology, stimulus, connectivity, precision, and output requirements across AxoSim and a detailed engine. Together, these analyses would strengthen both the scientific validity boundary and the practical interpretation of the runtime frontier.

12 Conclusion

AxoSim combines learned single-neuron fidelity with causal sparse population execution and differentiation through neuron- and contact-associated state. These capabilities are usually evaluated separately. Its strongest accuracy result is spike timing, accompanied by lower Voltage and Dynamics SERA than released Branch-ELM references. Paired synaptic interventions also yield lower teacher-response error across all compared model sizes. At population scale, a complete million-neuron step runs in real time at 362 contacts per neuron. A 65,536-neuron population remains 3.38 times faster than real time at 4,000 contacts, while complete full-BPTT updates remain real time for 20,000 neurons with 2,000 contacts per neuron.

The results also yield a methodological conclusion. Held-out trace gains and connected-population validity form two distinct levels of evidence because recurrent predictions change their own future inputs. AxoSim makes that distinction measurable by preserving contact semantics, exposing the ownership of adaptive state, and attaching each result to its measurement conditions.

Code and documentation. Installation, model and population interfaces, inference-time adaptation, worked examples, and the API reference are available at <https://axo.axym.org>.

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A Measurement contracts and supplementary evidence

A.1 Evaluation partitions

Table 6 gives the disjoint fitting, development, and final-evaluation units used by the principal comparisons.

Table 6: Data used for parameter fitting, schedule selection, and final evaluation.

Comparison	Fitting or adaptation	Development	Final evaluation
Ordinary AxoBench	training corpus	120 balanced traces	next 120: 24 threshold, 96 scored
Intervention response	fixed models	50 baseline calibration pairs	350 paired histories, 175 shards
Population specialization	64 recurrent contexts	16 recurrent contexts	20 untouched contexts
SHD adaptation	160 examples per source	80 examples on one source	800 examples on six sources
Connected populations	training traces	development connectomes	three paired unseen connectomes

A.2 Runtime measurement contracts

Table 7: Work included in each reported runtime quantity.

Quantity	Horizon and statistic	Included work	Excluded work
Isolated inference	500 steps, repeated median	model forward	data loading
Forward and backward	batch 32, 500 steps, 20 repeats	zero-grad, forward, loss, backward	loader, optimizer
Population inference	P4 advances, end-to-end p90	firing, routing, state, efficacy, 1% telemetry	data generation, initialization
Population adaptation	256 steps, seven replays	transfer, sparse input, forward, full BPTT, clip, Adam	graph capture

A.3 Model dimensions and optimization trajectories

The four full models use the dimensions below, which were fixed before the final training run.

Table 8: Temporal-core dimensions of the retained full models.

Model	Model width	State width	Layers	Head width
GRU Small	32	8	2	32
GRU Medium	128	32	2	128
Mamba Small	32	8	2	32
Mamba Medium	176	44	4	88

Training uses AdamW with batch size 32, initial learning rate 6×10^{-4} , cosine decay, and weight decay 10^{-4} . The objective combines scaled somatic-voltage MSE with weight 0.75, spike binary cross-entropy with weight 0.25, a direct SERA surrogate with weight 0.01, a differentiable F1 tolerance-curve term over 0–5 steps with weight 0.05, and near-spike voltage error within five steps with weight 0.002. The SERA surrogate evaluates 101 relevance thresholds.

Trained states at 24, 32, and 40 million presentations are compared on 120 development traces. GRU Small is retained at 32 million presentations, whereas the other three profiles are retained at 40 million. A separate set of 120 morphology-balanced traces supplies the final evaluation block.

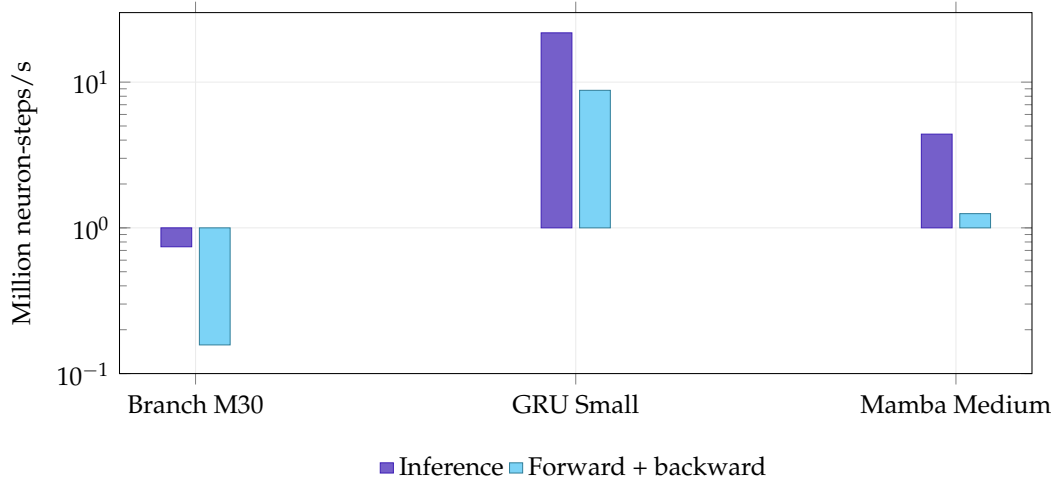


Figure 14: Paired inference and forward-plus-backward throughput for one representative of each architecture family. Each rate is reconstructed from the stored median latency, batch size, and 500-step window; loader and optimizer work are excluded.

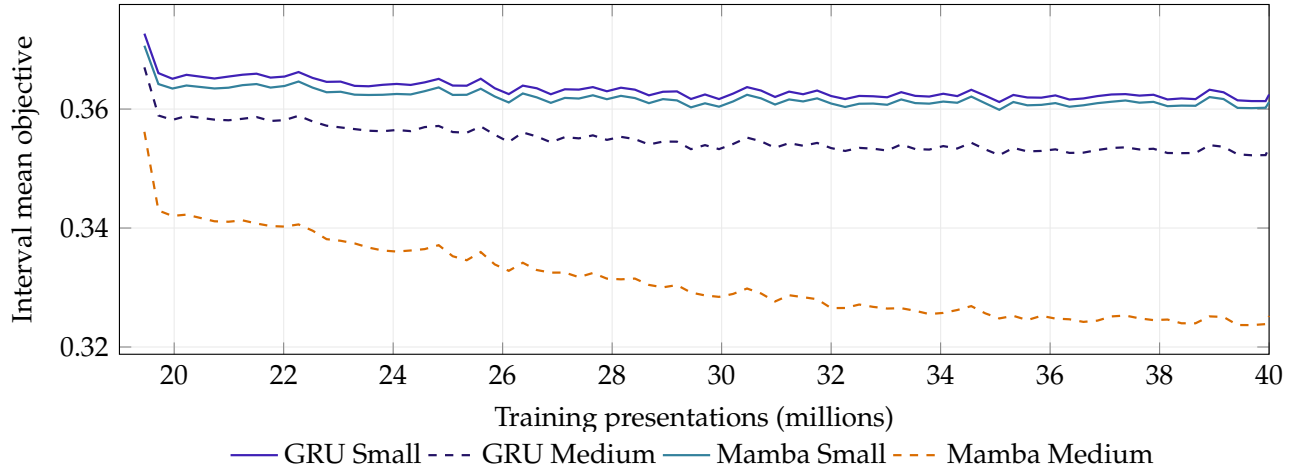


Figure 15: Interval mean optimization objective for the four stable models after the common training run resumed near 19.5 million presentations. The interval view distinguishes late plateauing and batch variation from numerical divergence.

A.4 Matched component ablations

The component comparison retraines ten two-layer Mamba configurations with 91,451 total parameters, 300,000 presentations and 9,375 updates per model, using seed 20260826 and the same five-morphology sampling schedule. Ablations disable one component while retaining its allocated parameters; the near-identity branch-saturation control freezes the affected transform parameters. The evaluation contains 120 traces, with 30 used for threshold calibration and 90 for scoring after the first 500 ms. Five hundred paired trace-bootstrap resamples estimate evaluation uncertainty for this common training seed. Table 9 reports the five components discussed in Section 2.

These records use Voltage Root-SERA in mV and the trapezoidal average of F1 across 0–5 ms. They therefore differ from the current Voltage SERA in mV^2 and arithmetic Mean F1. For every table column, positive values favor the retained component. Paired intervals for F1 at 5 ms include zero for morphology embedding and branch saturation, whereas their Voltage Root-SERA and trapezoidal-F1 intervals favor retention.

Table 9: Component-removal penalties after matched retraining. Entries give the point estimate and paired 95% interval. Voltage penalties are ablated minus full; F1 losses are full minus ablated.

Removed component	Voltage Root-SERA (mV)	Trapezoidal F1	F1 @ 5 ms
Route conditioning	.423 [.370, .477]	.087 [.072, .102]	.081 [.063, .099]
Morphology embedding	.207 [.162, .256]	.017 [.003, .032]	.008 [−.009, .025]
Local temporal refinement	.125 [.090, .163]	.016 [.003, .028]	.023 [.008, .038]
Branch saturation	.105 [.090, .123]	.009 [.002, .017]	.010 [−.001, .020]
Spike–voltage coupling	.091 [.066, .116]	.016 [.005, .027]	.017 [.004, .031]

The matched configuration uses width 58, state width 23, two temporal layers and 118 branches, with shared synaptic gain, bilinear interaction and a soma gate. Its readout differs from the separate heads used by the reported full models. A further 88,858-parameter configuration jointly removes these three interactions and uses separate heads; at batch 32 it is 13.0% faster, with a 0.021 mV increase in Voltage Root-SERA and a 0.003 increase in trapezoidal F1. This combined change informs the streamlined parameterization; the individual components’ necessity remains conditional on the readout and training configuration.

A.5 Compact adaptive coefficients

For token width D , recurrent width H , horizon P and output width O , the compact recurrence uses $Q = 2D + 3H + 2PO$ coefficients. Table 10 defines their packed groups. With $\tilde{u} = (u + o_u) \odot (1 + g_u)$, its exact transition and readout are

$$\begin{aligned}
d &= \sigma(\lambda + \delta_d), \\
p &= \tanh(W_u \tilde{u} + W_h h + b_h + o_h) \odot (1 + g_h), \\
h' &= d \odot h + (1 - d) \odot p, \\
\hat{y} &= (W_o h' + b_o + W_s \tilde{u}) \odot (1 + g_o) + o_o.
\end{aligned}$$

Here λ is the shared decay logit; $\hat{y}$ is reshaped into P native predictions with O coordinates each.

Table 10: The seven deployed coefficient groups. The measured compact model uses $D = 24, H = 8, P = 4, O = 2$, giving $Q = 88$.

Coefficient	Count	Symbol	Role
Token offset	D	o_u	input offset
Token gain	D	g_u	input sensitivity
Decay shift	H	δ_d	memory timescale
Proposal gain	H	g_h	recurrent proposal amplitude
Proposal offset	H	o_h	recurrent proposal offset
Output gain	PO	g_o	spike/voltage readout scale
Output offset	PO	o_o	spike/voltage readout offset

For neuron i of structural class $c(i)$, the coefficient vector is $q_i = a_i + m_{c(i)}$, where a_i is the behavior bank and m_c is the class bank. This additive parameterization permits selective optimization at either scale. Adding a vector to one class row and subtracting it from all corresponding neuron rows leaves the response unchanged, so unique attribution requires an additional constraint. Contact parameters instead act through positive efficacies $\exp(s_{i,k})$ before input aggregation. These coordinates describe predictive response changes; biological interpretation rests on their performance under controlled transfer.

A.6 Contact and neuron memory

Let K be incoming contacts, B branch features, R delay-ring slots, H recurrent coordinates and Q adaptive coefficients. If b_s, b_a, b_c are the bytes used for an efficacy, activation and coefficient, respectively, the principal persistent banks require $Kb_s + Rb_a + Hb_a + Qb_c$ bytes per neuron, plus Qb_c per structural class. The measured ring has $R = 8, B = 118$; BF16 ring entries occupy 1,888 bytes per neuron and the $H = 8$ recurrent state occupies 16 bytes. This accounting excludes shared weights, forecast buffers, routing workspace, telemetry and allocator overhead.

Table 11: Bank-level memory costs for $K = 4,000, B = 118, R = 8, H = 8, Q = 88$. The coefficient examples use BF16; contact deployment uses int8, while the training row includes FP32 values, gradients and two Adam moments. These are bank costs, rather than total GPU allocation.

Bank	Bytes	Example
Deployed contact efficacies	K per neuron	4,000
Recurrent state	$2H$ per neuron	16
Delayed branch ring	$2RB$ per neuron	1,888
Neuron coefficients (BF16)	$2Q$ per neuron	176
Class coefficients (BF16)	$2Q$ per class	176
Contact values, gradients and Adam state	$16K$ per neuron	64,000

Deployed efficacies are represented as signed 8-bit deviations from a positive baseline with a shared scale, requiring K bytes per neuron. Floating-point adaptation uses a separate representation. With FP32 efficacies, gradients and two Adam moment buffers, contact-associated parameter and optimizer storage is $16K$ bytes per neuron, before temporal activations. Thus 2,000 contacts contribute 32,000 bytes per neuron in this training accounting. Sparse event accumulation removes the dense $N \times T \times K$ input history, while full temporal BPTT retains the activations needed across the supplied T -step sequence.

A.7 Native-time population contracts

The compact forecast consumes exactly one completed $P = 4$ input patch. The first four native samples are a startup prefix with no valid forecast and no emitted recurrent events. Thereafter each output has its own native timestamp. Every retained contact is validated to have delay at least 4 ms, and the delay ring covers both the minimum delay and forecast horizon. Neuron phases are assigned independently of morphology and source role within stratified tiles.

The stateful population interface maintains neuron state, the delayed queue and mutable contact efficacies across calls. Supplied-history tensor execution provides the differentiated sequence, whereas the procedural phase/tile-major implementation supplies the measured large-population inference path. General explicit-contact graphs and the structured procedural topology share contact semantics but have distinct implementations and runtime measurements. Floating-point efficacy in the general interface is separate from the quantized efficacy bank used by the deployed timing profile.

A.8 Complete-update capture parity

The eager/captured SHD comparison performs 40 updates from identical source parameters and optimizer state on the same ordered batches. Both paths use capturable Adam, the same learning rate, cross-entropy objective, adaptive parameters and clipping rule. Capture warms up on a side stream, records the complete update, and restores the model and optimizer states before replay. The maximum final parameter discrepancy is zero. Eager execution takes 6.404 s, replay 1.836 s and separate capture preparation 0.737 s. These times establish parity and replay acceleration for this workload; the dense-contact full-BPTT timings use their own measurement contract.

A.9 Matched intervention-response comparison

The comparison selects the first two pairs from each of 200 dataset shards, using pair ordinal alone. Five interventions and five morphologies define 25 strata, with eight shards per stratum. The first shard of every stratum supplies calibration, giving 50 pairs; the remaining seven supply 350 scoring pairs in 175 shard clusters. Selection is independent of model predictions and preserves equal condition and morphology counts.

Each model’s single prevalence-matched spike threshold is calibrated from baseline predictions and baseline teacher spikes only, then frozen for both members of every scoring pair. The four full AxoSim and five released Branch-ELM models retain their selected weights. Each six-second history is evaluated from a fresh initial state at 1-ms resolution, with the first 500 ms excluded. There are 70 scoring pairs and 385 scored neuron-seconds per intervention condition.

The NeuronIO voltage coordinate is $\hat{v} = 0.1(v + 67.7)$, so predictions are converted once by division by 0.1 and subtraction of 67.7 mV. Fidelity uses raw, unclipped teacher soma voltage. Training targets are clipped at -55 mV; their consistency check is $\text{target}/0.1 - 67.7 = \min(v_{\text{raw}}, -55)$. This establishes the saved target relation while preserving the raw-teacher metric policy.

For each pair, let $\Delta v = v^{\text{intervention}} - v^{\text{baseline}}$. Response SERA applies the conditional squared-error integral to $\Delta \hat{v} - \Delta v$, with relevance equal to the maximum of baseline-voltage, intervention-voltage and teacher-response-tail relevance. Counts, squared errors and cross-products are pooled before computing the summary metrics. The zero-response control sets $\Delta \hat{v} = 0$. Centered Pearson correlation describes response-shape association, while the least-squares gain $\sum \Delta \hat{v} \Delta v / \sum (\Delta v)^2$ describes the projection of the predicted change onto the teacher change. The uncentered response cosine is retained separately in the numerical records.

One thousand paired bootstrap draws resample shards within each of the 25 strata, retaining the two dependent traces within each selected shard. Every draw recomputes pooled sufficient statistics for all nine models. The primary family contains all twenty AxoSim–Branch-ELM response differences. A maximum absolute standardized bootstrap statistic provides simultaneous 95% intervals across this family, with critical value 2.478. For example, GRU Small minus Branch-ELM M1 is -30.62 mV^2 , with simultaneous interval $[-32.47, -28.78]$. All twenty intervals are below zero. These intervals quantify shard-sampling uncertainty for the fixed models and morphologies. Table 12 gives individual model intervals, distinct from these paired family intervals. Figure 6 shows descriptive pooled firing-rate changes, without interpolation between the unordered intervention categories.

Table 12: Absolute fidelity and response diagnostics on the 350 scoring intervention pairs. Response SERA is in mV^2 , with individual 95% bootstrap intervals; the zero-response control gives 298.27. Pearson and gain use the same raw-teacher voltage changes.

Model	Response SERA	95% interval	Mean F1	Pearson	Gain
AxoSim–GRU Small	255.55	[248.44, 262.29]	.439	.487	.276
AxoSim–GRU Medium	256.30	[249.19, 263.30]	.470	.485	.269
AxoSim–Mamba Small	256.31	[249.31, 263.11]	.416	.483	.271
AxoSim–Mamba Medium	258.78	[251.47, 265.76]	.491	.485	.303
Branch-ELM M1	286.17	[278.71, 293.70]	.070	.336	.183
Branch-ELM M10	287.85	[280.38, 295.27]	.062	.331	.179
Branch-ELM M30	287.67	[280.15, 295.10]	.076	.333	.183
Branch-ELM M50	287.90	[280.35, 295.34]	.073	.333	.186
Branch-ELM M100	287.92	[280.41, 295.35]	.075	.332	.186

The teacher-input mapping assigns synaptic channels to dendritic sites using normalized arbor length. Site silencing removes a declared band of input channels, so its physical interpretation is a perturbation of synaptic drive at mapped sites. Event removal and sign-selective removal alter the corresponding input events; jitter alters their times. The response results therefore concern these five input interventions, with the same teacher mapping used for baseline and intervention.

A.10 Intervention-response geometry

The intervention analysis uses the final AxoSim–Mamba Medium model and all 12,500 paired traces in axobench–interventions. The five intervention families are event dropout, excitatory dropout, inhibitory dropout, site silencing, and temporal jitter, crossed with five morphologies and eight dataset shards per combination. Baseline state is subtracted from intervention state for every pair. The first 500 ms are excluded, and the remaining 5,500 ms are averaged into eleven consecutive windows.

The signal column in Figure 2 uses the 500-ms window with the highest teacher spike count among one deterministically selected pair from each of the 200 shards. This trace exposes measured input, state, and output signals in the architecture diagram. It is excluded from every reported aggregate statistic. The input display uses 10-ms event counts, while the temporal representation, voltage, and spike probability are sampled at the native 1-ms resolution. The displayed probability is the sigmoid of each individual logit. PC 1 is computed after standardizing each temporal feature within this window.

Panel (a) groups each shard deterministically into disjoint mini-cohorts of seven or eight pairs, producing 1,575 points. Each point concatenates the mean 176-dimensional temporal-state displacement in the eleven windows. Features are standardized before fitting a three-dimensional cosine-distance

UMAP with 30 neighbors, minimum distance 0.12, and random seed 20260930. UMAP coordinates are used only for the display.

Panel (b) selects one pair from each shard by hashing the seed, intervention, and shard name. State displacements are averaged into 500-ms windows, and one PCA is fitted across every selected trace and time. The first three components explain 53.2%, 21.1%, and 8.9% of the variance. Condition paths average 40 selected traces. For the gray flow field, each individual transition is assigned to a cell in a $4 \times 4 \times 4$ grid spanning the 3rd–97th percentile range. Arrows report mean displacement in occupied cells with at least eight transitions; the 32 most populated cells are shown.

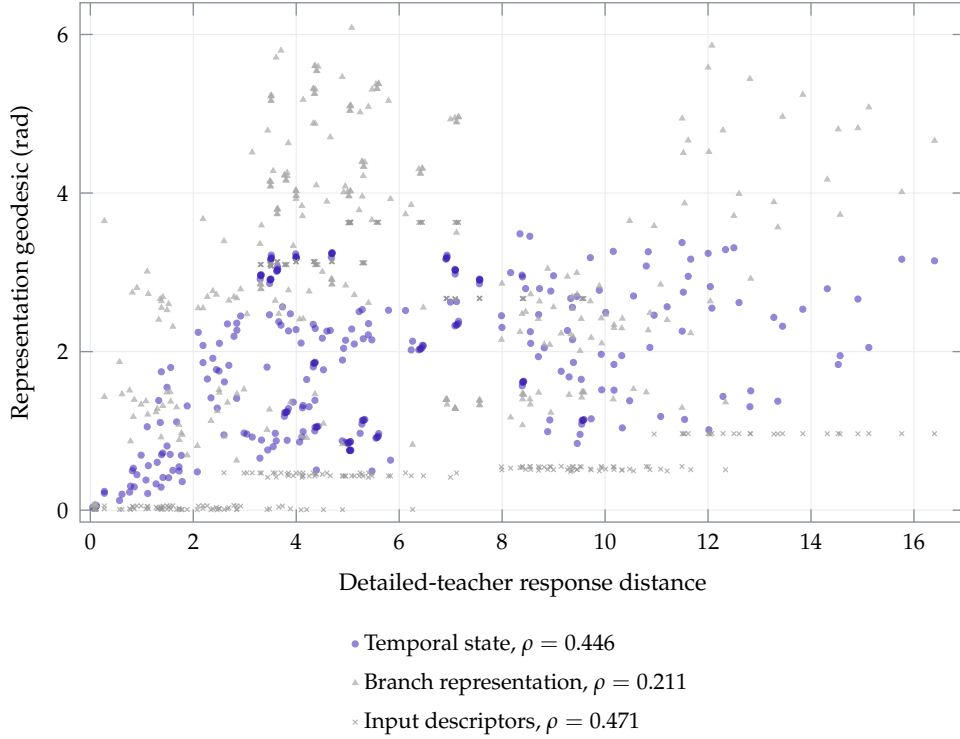


Figure 16: Association between teacher-response and representation distances across 25 intervention-morphology groups, or 300 group pairs. Geodesics use each original standardized representation; teacher distances use standardized voltage and spike-response signatures. Spearman correlations are 0.446 for temporal state, 0.211 for branch features and 0.471 for input descriptors. The input control relates the observed response organization to information in the imposed perturbation.

The external-ruler comparison averages the eight shards within each of the 25 groups defined by intervention and morphology. In each representation, feature-standardized angular distances form the smallest connected symmetric nearest-neighbor graph. The required neighbor counts are five for branch state, six for temporal state, and five for the input descriptor. Shortest-path distances are then compared by Spearman correlation with Euclidean distance between standardized detailed-teacher signatures. Each teacher signature contains eleven mean voltage changes and eleven spike-count changes. The 95% intervals come from 1,000 resamples of the eight shards within every group; each resample reconstructs the group representations and graph. The direction-consistency margin subtracts the mean cosine similarity across interventions within a morphology from the mean similarity across morphologies within an intervention.

The direction-consistency margin is 0.563, with a 95% interval of [0.559, 0.566]. Teacher-distance Spearman correlations and their intervals are 0.446 [0.443, 0.451] for temporal state, 0.211 [0.167, 0.213] for branch features and 0.471 [0.443, 0.498] for input descriptors.

A.11 Central figure values and axis mappings

Table 13 gives the values represented in Figure 1. Inference throughput uses the absolute mapping $1 - \exp(-r/10^7)$, with r in neuron-steps/s. Voltage and Dynamics SERA use $\exp(-E/350)$ and $\exp(-E_\Delta/450)$, respectively, so zero error maps to one and unbounded error approaches zero. Mean F1 and F1 at 5 ms are untransformed. These mappings are fixed independently of the compared values. The Branch-ELM panel shows M1, M30, and M100, covering the smallest released model, its highest Mean F1, and the largest released model. All five sizes use the same evaluation and timing contracts; M10 and M50 are listed below rather than overlaid on their almost coincident polygons.

Table 13: Raw values used by the central figure. Rates are millions of neuron-steps/s; Voltage SERA is in mV², and Dynamics SERA is in mV²/ms².

Variant	Inference	Voltage SERA	Dynamics SERA	Mean F1	F1 at 5 ms
CoreNEURON reference	0.00085	0	0	1.000	1.000
Branch-ELM M1	0.788	312.26	448.64	0.053	0.070
Branch-ELM M10	0.717	309.76	448.60	0.066	0.088
Branch-ELM M30	0.742	307.41	448.57	0.077	0.102
Branch-ELM M50	0.735	307.18	448.58	0.076	0.101
Branch-ELM M100	0.712	307.34	448.56	0.073	0.098
AxoSim-Mamba Medium	4.396	244.46	439.00	0.520	0.641
AxoSim-GRU Small	21.786	243.92	429.12	0.486	0.604

CoreNEURON supplies the target trajectories, so its self-comparison has zero SERA and unit F1. The simulation rate comes from a different hardware and workload record and appears as an absolute measurement rather than the denominator of a cross-hardware speedup. Learned-model rates use the shared 500-step inference contract described in Section 5.

The learned-model comparison concerns fitted quality and measured execution rates. AxoSim and the released Branch-ELM models received different upstream optimization budgets, so these measurements do not establish relative training efficiency.

A.12 Pareto-arrow selection and FLOP contract

The orange arrow in each panel of Figure 4 follows one fixed rule. Within the Branch-ELM group and the combined AxoSim group, points dominated on both the displayed efficiency and fidelity axes are discarded. The remaining coordinates are min-max normalized within their group after orienting both axes so that larger values are better. The selected point is the nondominated candidate with the smallest Euclidean distance to the ideal corner ((1,1)), with model identifier as a deterministic tie-breaker. The precomputed endpoint table in `data/tikz/pareto-arrows.csv` decouples selection from figure layout.

Forward-pass FLOPs count the operations recognized by `torch.profiler` for one batch-one, 500-step sequence with 1,278 input features. The same measurement is applied to every learned model. Parameter counts include trainable parameters only. Neither quantity includes data loading or optimizer operations.

A.13 Population timing and connectivity

Both population measurements use one RTX 5090 and BF16 compact-neuron execution at 1 ms resolution. Each timed step includes delayed routing that preserves source identity, mutable positive efficacies, recurrent state, and pinned-host voltage telemetry for 1% of neurons. The million-neuron measurement uses closed-loop spike discovery and 362 retained contacts per neuron. Three target-activity replicates span 1.174–1.197 billion neuron-steps/s, with a median of 1.177 billion neuron-steps/s.

Figure 8 uses 65,536 neurons to measure denser graphs at 100, 200, 400, 800, 1,600, 2,000, 3,200, 4,000, 6,400, 8,000, and 10,000 retained recurrent contacts per neuron. A threshold scan remains in the timed step, after which source events are replayed at fixed morphology- and role-conditioned target rates. This replay holds activity constant while the measurement varies routing density. External input width remains fixed at 1,278 afferents, and every recurrent contact retains an independently mutable efficacy. Repeated source–target pairs are permitted, as multiple synapses may connect the same neuron pair.

Each contact count is measured on three independently realized graphs. The reported value is the median p90 step rate, with the full replicate range shown by the error bars. In this fixed-activity routing sweep, the 4,000-contact median is 249.87 million neuron-steps/s and the 10,000-contact median is 142.09 million neuron-steps/s, equivalent to 2.17 times real time. The curve connects measured medians and terminates at the 10,000-contact endpoint.

The orange 4,000-contact marker replaces source replay with threshold-derived closed-loop firing. Across three graph realizations, the measured-to-target rate ratio spans 1.021–1.026 with fixed thresholds throughout the measured window. The median p90 rate is 221.47 million neuron-steps/s, equivalent to 3.38 times real time for 65,536 neurons.

Quantitative mouse-neocortex estimates are on the order of thousands of contacts per neuron, with a reported average near 8,200 and substantial region and cell-type variation. Contacts also differ from counts of unique presynaptic partners. Population size, connectivity, and simulated time per wall clock second should therefore be reported together.

A 426-contact implementation used a stride-three bit-mask selector. Bit-mask reduction is valid only for power-of-two moduli; with a non-power-of-two fanout it collapses residue classes and does not realize the declared destinations. Correct modulo routing restores every class and yields more scattered target addresses. In the directly comparable records, median routing time rises from 1.579 to 1.965 ms. The increase is consistent with both the arithmetic remainder and the less-local target addresses; the timing record does not identify their individual contributions.

A.14 Population-input specialization details

Table 14: Target-domain metrics after 30,000 population-context presentations. SERA values use the same squared units as the ordinary AxoBench metrics.

Model	Mean F1	Voltage SERA	Dynamics SERA
AxoSim-GRU Small	0.379	48.10	20.24
AxoSim-GRU Medium	0.402	50.34	23.49
AxoSim-Mamba Small	0.364	47.89	21.34
AxoSim-Mamba Medium	0.412	49.23	24.23
Branch-ELM M50	0.132	36.87	16.36
Branch-ELM M100	0.136	36.93	16.36

Recurrent-domain specialization complements the ordinary-domain comparison. Section 6 gives the direct cross-family SERA result on ordinary AxoBench. The target-domain analysis instead measures how each model adapts its spike behavior to recurrently generated inputs.

A.15 SHD contact-density analysis

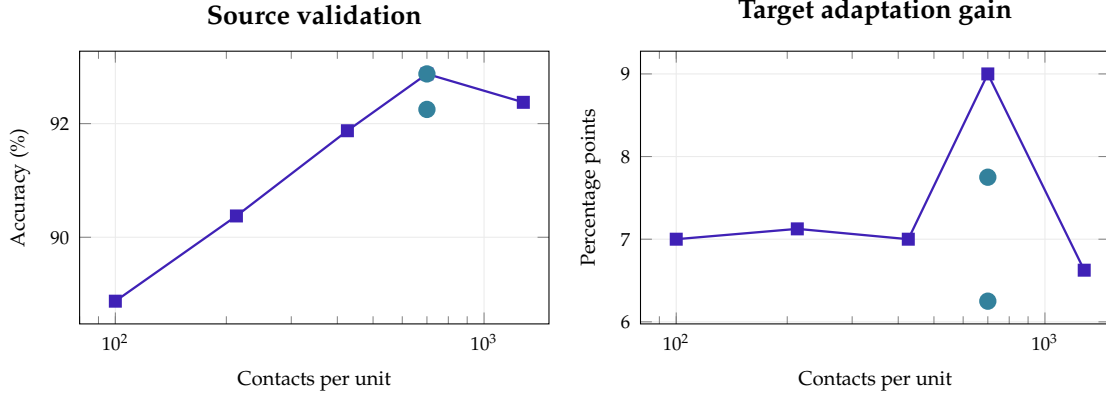


Figure 17: SHD adaptation as the number of synaptic contacts changes. The development result on one source is non-monotonic; two independently trained sources confirm the selected 700-contact setting.

A.16 Connected diagnostic definitions

Let $r_{s,m}$ denote the firing rate of excitatory morphology group m over either the full or tail window s , and let r_m^* denote its contract target. The morphology discrepancy is

$$D_{\text{morph}} = \frac{1}{2|\mathcal{M}|} \sum_{s \in \{\text{full}, \text{tail}\}} \sum_{m \in \mathcal{M}} \left[\log \left(\frac{r_{s,m}}{r_m^*} \right) \right]^2.$$

For per-neuron input-count lag correlations $\rho_{s,\ell}$, evaluated at $\ell \in \{1, 4, 16, 64, 256\}$ ms, the temporal discrepancy is

$$D_{\text{temp}} = \frac{1}{2|\mathcal{L}|} \sum_s \sum_{\ell \in \mathcal{L}} \left(\frac{\rho_{s,\ell} - \rho_\ell^*}{0.2} \right)^2.$$

Finally, let q range over global rate, excitatory rate, inhibitory rate, and input-count coefficient of variation. The stationarity discrepancy is

$$D_{\text{stat}} = \frac{1}{4} \sum_q \left[\log \left(\frac{q_{\text{tail}}}{q_{\text{full}}} \right) \right]^2.$$

Targets and normalizations are fixed before held-out testing. The three quantities remain separate throughout evaluation.

A.17 Conditional single-neuron examples

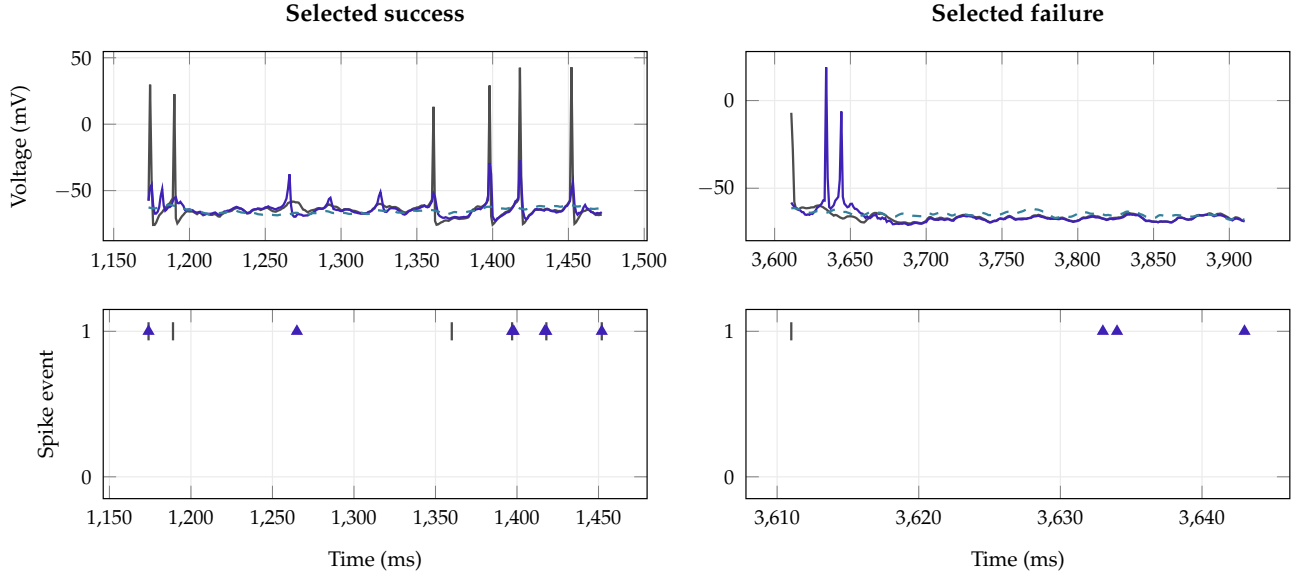


Figure 18: Balanced conditional examples from the final comparison block. The upper row is the spike-containing 300 ms window with the largest voltage-MSE advantage for the selected AxoSim-Mamba model over the strongest Branch-ELM spike baseline; the lower row is the largest disadvantage under the same rule. These selected examples expose one success and one failure and are not estimates of average performance. Black denotes the detailed teacher, solid purple AxoSim, and dashed cyan Branch-ELM.

A.18 Inference and full-BPTT contact memory

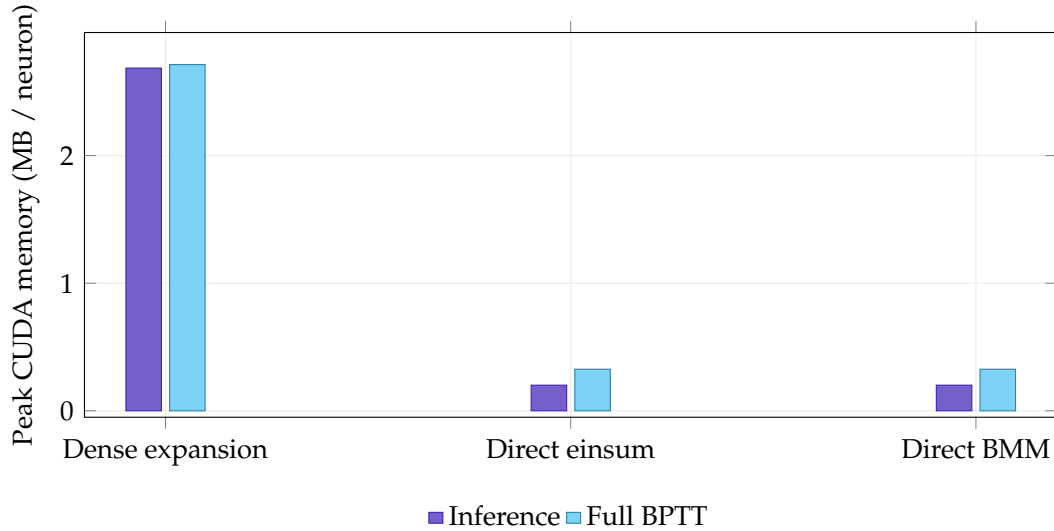


Figure 19: Peak CUDA memory per neuron for inference and full BPTT across dense branch expansion, direct einsum, and direct BMM contact-input implementations at 24 contacts and 256 timesteps.

Dense branch expansion uses approximately 2.686 MB per neuron for inference and 2.713 MB for full BPTT. Direct einsum and direct BMM each use approximately 0.201 MB for inference and 0.326 MB

for full BPTT. Normalized per-neuron memory permits comparison across the different executable population sizes; the source CSV also records raw populations and byte counts.

A.19 Adaptation timing boundary

The dense-contact benchmark optimizes behavior, morphology, and synaptic parameters together through 256 native-millisecond steps. At 2,000 retained contacts per neuron, 20,000 neurons pass all seven changing updates and 21,000 exceed the 256-ms deadline. At 4,000 contacts, 10,000 neurons pass and 11,000 exceed the deadline. The reported curve uses the slowest update at each population size. Timing includes fresh pinned-host input, exact sparse contact aggregation, transfer, forward execution, loss, full backward propagation, gradient clipping, and Adam; one-time graph capture is excluded.